

Morphological study Gennaker offshore wind farm



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Summary

The Gennaker Offshore Wind Farm (OWF) is a proposed development in the Baltic Sea, approximately 15 km of the coast of Germany. The project is being developed by OWP Gennaker GmbH and managed by Skyborn Renewables. The project area covers approximately 88 km². Within this area three subareas are allocated for the installation of 103 Wind turbine Generators (WTG) and two Offshore Substations (OSS). The objectives of the seabed mobility study are to assess the morphodynamics and provide an estimation of the future seabed levels in and around the Gennaker OWF. To this end, a data-driven analysis is performed that consist of three parts: overview of the characteristics in the Gennaker OWF, morphodynamic analysis and estimation of future bed levels.

The Baltic Sea is characterized by mild hydrodynamic conditions. In the region of the Gennaker OWF, tidal amplitudes reach up to 0.15 – 0.20 m, with mild tidal current velocities, reaching depth-averaged velocities mostly below 0.1 m/s. Residual flows result in total velocities typically below 0.3 m/s. Significant wave heights at the OWF are generally below 1.5 m. The seabed in the Gennaker OWF can be characterised as a sloped seabed with local minor undulations. Bed level in the area ranges from -14 to -20 m relative to MSL. Close to the Gennaker OWF two areas with rhythmic bedforms are observed. These bedforms are mainly located in bed levels above -11 m relative to MSL.

Several areas with smaller notable features are observed in and around the Gennaker OWF. Natural features include megaripples located just south of the most western point of the OWF and sediment structures (local small-scale surficial sediment deposits of significantly coarser material) spread throughout the area. Manmade structures include two cable repairs located along the high voltage cable that travels through the OWF. The surficial sediment in and around the OWF is predominantly classified by very fine sand in the northeast and fine sand in the southwest. Locally and to the east of the OWF medium to very coarse sand and gravel is present.

The morphodynamic data-analysis indicates that over the period 2013 to 2022 the mobility of the seabed in the Gennaker OWF is limited. After correction of the bathymetries (i.e., to correct for differences in vertical reference levels) the mobility of the seabed is estimated at maximum 0.15 m vertical change over a period of 5 to 7-years (i.e., from 2013 and 2015 to 2022). Analysis of notable areas showed that largest seabed changes in the area (i.e., approximately ± 0.30 m) occur due to manmade interventions (cable repairs). The observed changes in seabed levels which can be attributed to natural morphodynamic activities are predominantly governed by the migration of small scale bedforms such as megaripples and suggested around sediment structures (although the latter may also be associated with data inaccuracies).

To assess the impact of extreme events on seabed levels in Gennaker OWF numerical modelling was performed. A coupled hydrodynamic and wave model together with a morphodynamic module computing bed level changes during an extreme event was used. In this study the Anatol storm was considered, which, modelled, roughly represents an event with a return period of 50 years. Results indicate that seabed level changes in Gennaker OWF are expected to be limited (< 0.07 m) during such an extreme event.

To account for seabed changes caused by small-scale dynamics (i.e., migration of megaripples) and large-scale dynamics (i.e., regional seabed changes) we thus consider a uniform uncertainty of ± 0.01 m per year over the full extent of the Gennaker OWF. For the extreme events a uniform value of ± 0.07 m was considered, taking into account both the results

of the extreme event assessment and the recurrence of such an event. Together with the uncertainties related to multibeam measurement (± 0.20 m) a maximum bathymetric change of ± 0.66 m with respect to the Gennaker 2022 bathymetry is expected at any given location over the lifetime of the windfarm (i.e., 4 years of preparation and installation till 2025/2026 and 35 years thereafter).

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1 Introduction

1.1 Background

The Gennaker Offshore Wind Farm (OWF) is a proposed development in the Baltic Sea, approximately 15 km of the coast of Germany, see the maps in Figure 1.1. The project is being developed by OWP Gennaker GmbH (hereafter referred to as the client). The project area covers approximately 88 km². Within this area three subareas are allocated for the installation of 103 Wind turbine Generators (WTG) and two Offshore Substations (OSS). Moreover, the Gennaker OWF contains a 6 km² area that is occupied by the existing EnBW Baltic 1 OWF. Insight into the mobility of the seabed in and around the Gennaker OWF is important to determine if risks can be expected in the supply of generated wind energy, e.g., with respect to the stability of wind turbines and the exposure of power cables.

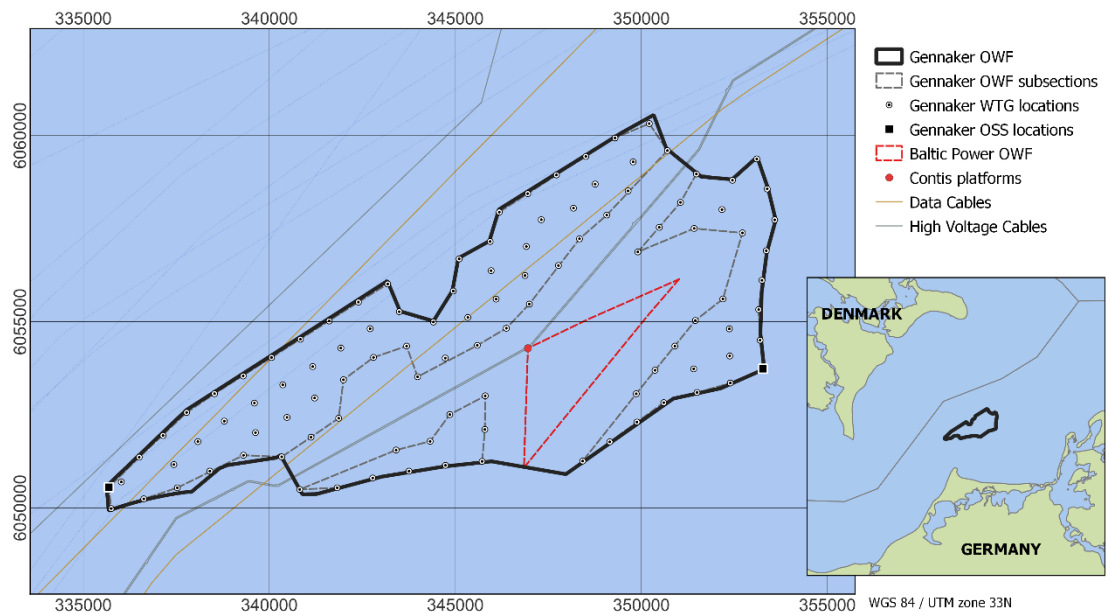


Figure 1.1 Map showing the contours of the Gennaker OWF (black lines), subareas of the Gennaker OWF (black dashed lines), existing EnBW Baltic 1 OWF (red dashed lines), data cables (brown lines) and high voltage cables (grey lines). The lower right panel shows the Gennaker OWF on a larger scale.

1.2 Objectives

The overall objective of the seabed mobility study is to assess the morphodynamics in and around the Gennaker OWF (see Figure 1.1). Furthermore, the objective is to provide an estimation of the future seabed levels in the area for a period of 35 years after installation in 2025/2026.

1.3 Methodology

To achieve the abovementioned objectives, a data-driven analysis is performed. An overview of the available data used in this morphodynamic assessment is presented in Chapter 2. The data-driven analysis and extreme event assessment is presented in Chapter 3 and consists of three parts: overview of the characteristics in the Gennaker OWF area, morphodynamic analysis and estimation of future bed levels.

Firstly, the available data is processed and reviewed. Thereafter, an overview of the hydrodynamic, wave, bathymetric and geotechnical and geophysical characteristics in and around the Gennaker OWF area is given.

The morphodynamic analysis will demonstrate to what extent (if present) bed features such as mega ripples, sand waves and sand banks are present in and around the Gennaker OWF. It is important to make this characterisation since these features change and migrate under different circumstances, hence determine the extent of the mobility of the seabed. Moreover, multiple bathymetry datasets are used to determine if and where the bathymetry changes occur. Thereafter, the observed bed features and bed level changes are validated using the available hydrodynamic, wave, geotechnical and geophysical data. Finally, a numerical modelling assessment was conducted to derive seabed level changes during an extreme event.

The results of the morphodynamic analysis are used to determine morphodynamic trends and uncertainties. Lastly, these trends and uncertainties are used to make an estimation of the future bed levels in and around the Gennaker OWF.

2 Overview of the available data

2.1 Introduction

This section contains an overview of the available data. The hydrodynamic and wave data is described in Section 2.2. Section 2.3 describes the available bathymetry data. Finally, the geotechnical and geophysical data are discussed in Section 2.4.

2.2 Hydrodynamic and wave data

The main source of hydrodynamic and wave data for this study is the metocean report for Gennaker OWF by Deltares (Deltares, 2022). In this report the hydrodynamic and wave conditions are given for three locations (i.e., 028, 101 and 140) around the Gennaker OWF. These locations are shown in Figure 2.1 (location 028) and Appendix A. The data, both normal and extreme conditions, at these locations is used to get an understanding of the hydrodynamic and wave characteristics at the Gennaker OWF and to assess the impact of an extreme event.

Figure 2.2 shows the roses of the depth-average total, tidal and residual current speed and direction for normal conditions at the three locations around the Gennaker OWF. The roses in this figure show that the depth-averaged currents are generally below 0.3 m/s and predominantly oriented towards the east-northeast followed by west-southwest. These currents are mainly made up of residual currents, since tidal current velocities are mostly below 0.1 m/s. Similarly, the roses for the significant wave height and direction are shown in Figure 2.3. This figure shows that significant wave heights are generally below 1.5 m (roughly 90% of the time) with associated peak wave periods below 7.0 s. The monthly roses are not included in this report but can be found in Deltares (2022). The monthly roses show that during the winter the waves generally approach from the west-southwest, while during summer the waves predominantly approach from the northeast.

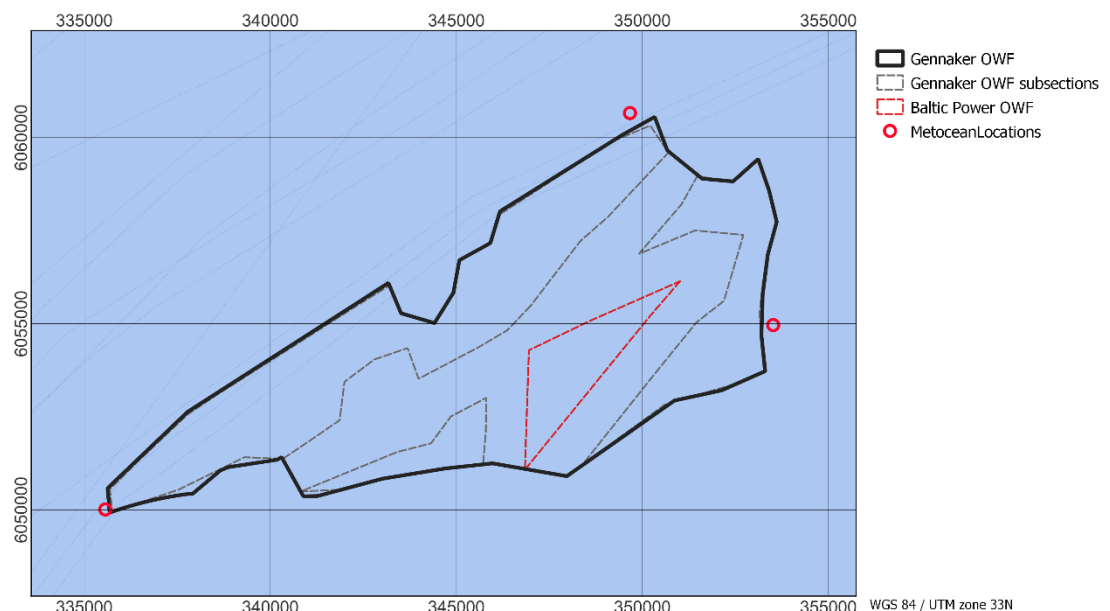


Figure 2.1 Metocean output locations (red circles) around the Gennaker OWF. (Deltares, 2022).

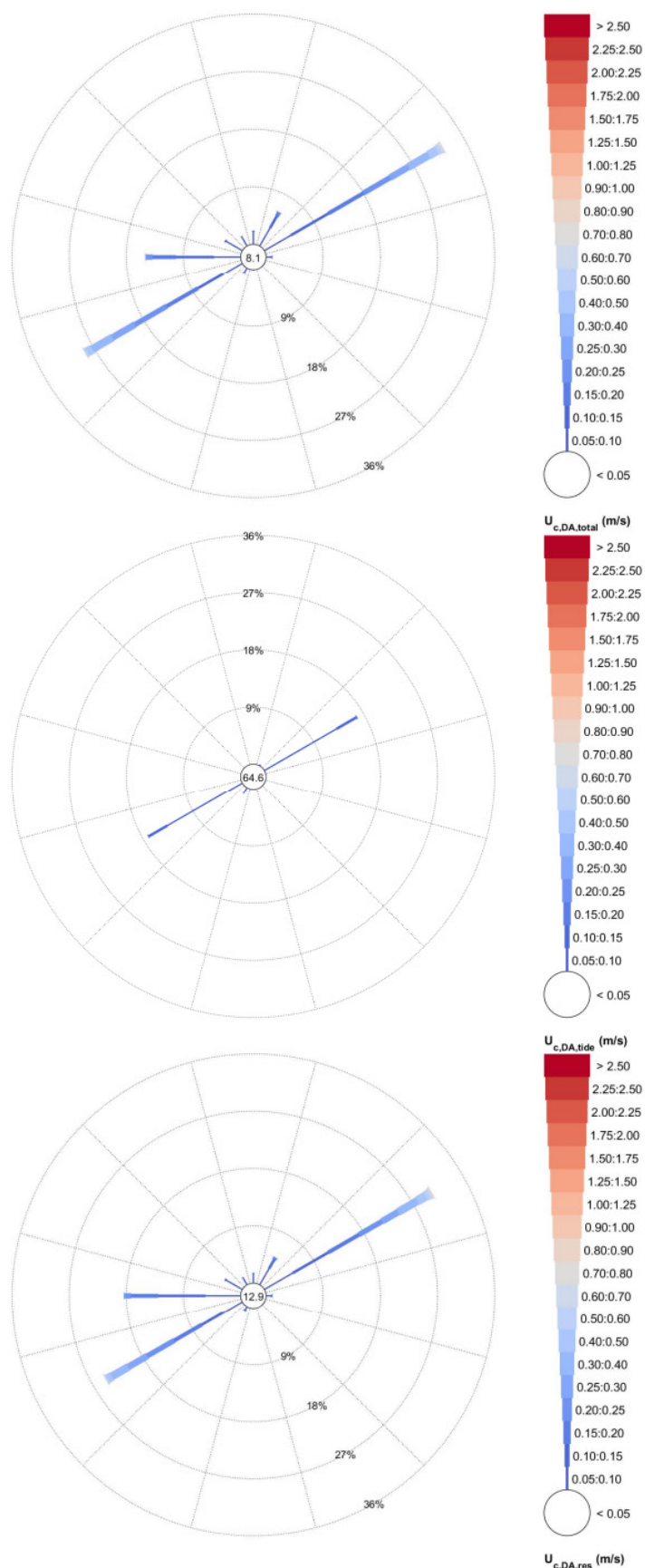


Figure 2.2 Roses of the depth-average total (top panel), tidal (centre panel) and residual (bottom panel) current speed and direction for normal conditions at location 028 (Deltares, 2022).

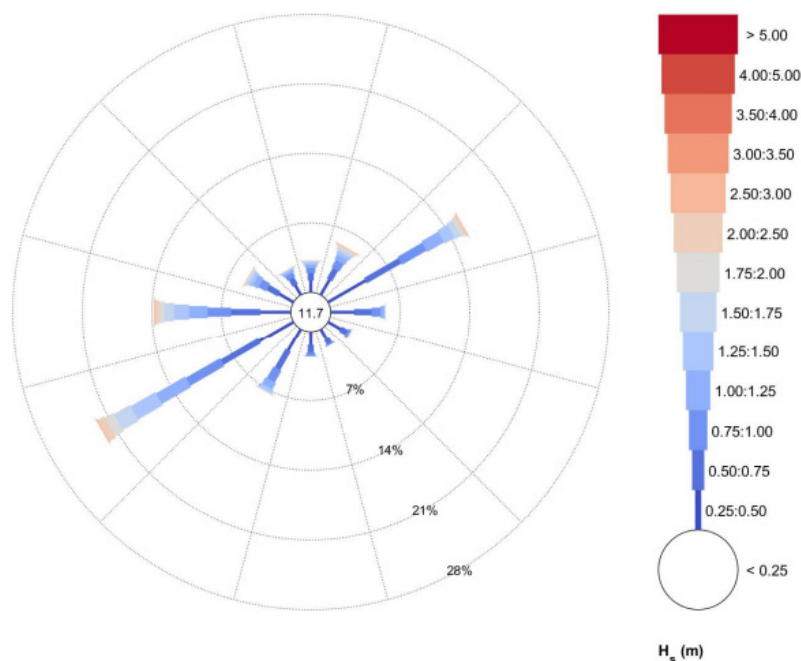


Figure 2.3 Roses for the significant wave height and direction for normal conditions at location 028 (Deltares, 2022).

2.3 Bathymetric data

The two main sources of bathymetric data for this study comprise of (i) three existing ‘historical’ bathymetrical datasets available from the client and (ii) one existing ‘historical’ dataset available from BSH (the Federal Maritime and Hydrographic Agency of Germany) (Vermessungsbüro Weigt, 2016; Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022; Vermessungsbüro Weigt, 2022). These bathymetries are used to determine the seabed characteristics and to estimate future seabed levels in and around the Gennaker OWF.

Table 2.1 contains detailed information on the four surveys, including measurement dates, vertical reference level, data density and resolution. It is noted that the vertical reference level for the 2013 and 2015 surveys (i.e., DHHN2016) differs from that of the other surveys (i.e., MSL). The DHHN2016 is equal to MSL - 0.11 metres (Vermessungsbüro Weigt, 2016).

Table 2.1 Historical bathymetrical surveys available in and around the Gennaker OWF. Data density is specified as Low (resolution of 10 m or larger) and High (data with a 1 by 1 m resolution or finer).

Survey ID	Survey Name	Survey date [mm-yyyy]	Vertical reference level	Data density [m]	Resolution [m]
Site specific bathymetry data – received from the client					
BAT_WPD_OWP_GEN_NW_Flaeche_UTM84_33N_DHHN_30cm (Vermessungsbüro Weigt, 2016)	Gennaker 2015 NW bathymetry	11-2015	DHHN2016*	High	0.30
BAT_WPD_OWP_GEN_SO_Flaeche_UTM84_33N_DHHN_30cm (Vermessungsbüro Weigt, 2016)	Gennaker 2013 SO bathymetry	11-2013	DHHN2016*	High	0.30
GEN-GEO-MBES-BAT survey 2022-gridded-25cm-25833-MSL (Vermessungsbüro Weigt, 2022)	Gennaker 2022 bathymetry	05-2022	MSL	High	0.25

Bathymetry data for wider area – obtained from various sources

Baltic Sea Digital Terrain Model (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022)	Baltic Sea composite bathymetry	**	MSL	Low	100 - 500
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* DHHN2016 is equal to MSL - 0.11 m

** Composite bathymetry created from surveys conducted between 1992 and 2016, inside Gennaker OWF the data is measured between 1992 and 2000.

Figure 2.4 shows the bathymetries for the two surveys conducted in 2013 and 2015 (i.e., Gennaker 2015 NW and 2013 SO). Together these surveys cover the full extent of the Gennaker OWF area, except for the area around the existing Baltic Power OWF. Figure 2.5 shows the bathymetry for the survey conducted in 2022. This survey also covers the full extent of the OWF, except for the area around the existing OWF. Finally, Figure 2.6 shows the composite bathymetry for the Baltic Sea. This bathymetry was created from surveys conducted between 1992 and 2016. The vertical reference level for all bathymetries has been converted to MSL.

Other data sources such as GEBCO and data from admiralty charts are not used in this morphodynamic analysis because of limited spatial resolution and coverage and large uncertainties in bed levels.

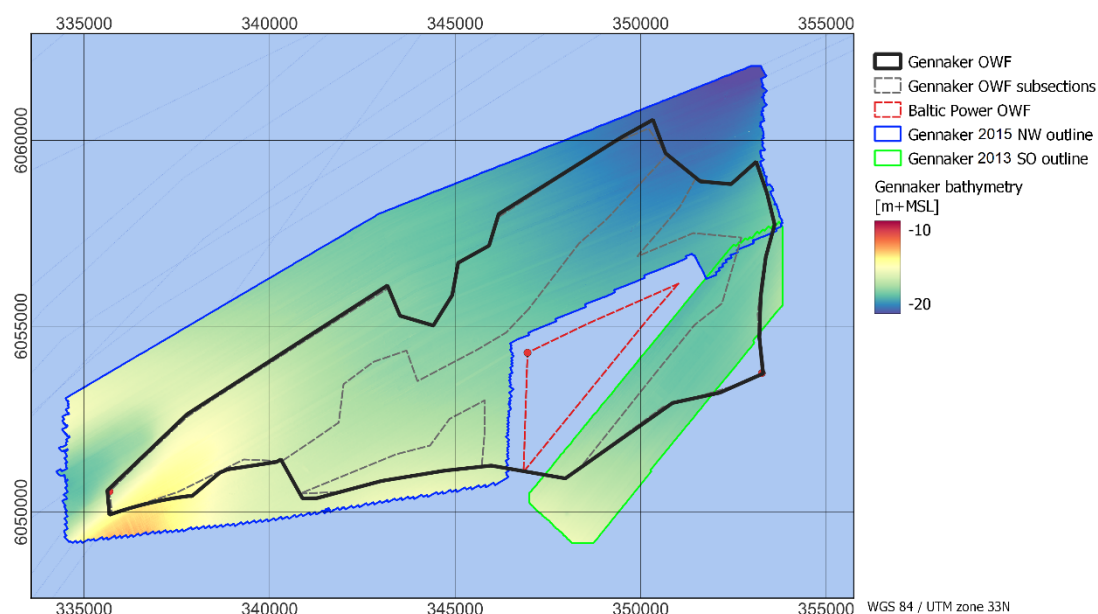


Figure 2.4 Gennaker 2015 NW (blue outline) and Gennaker 2013 SO (green outline) bathymetries [m+MSL].

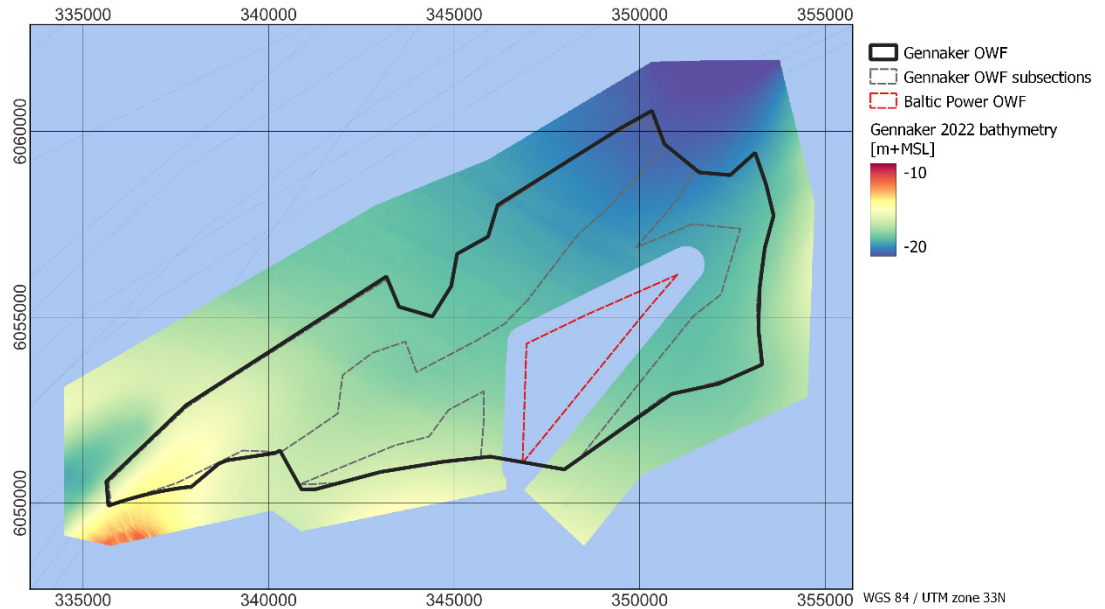


Figure 2.5 Gennaker 2022 bathymetry [m+MSL].

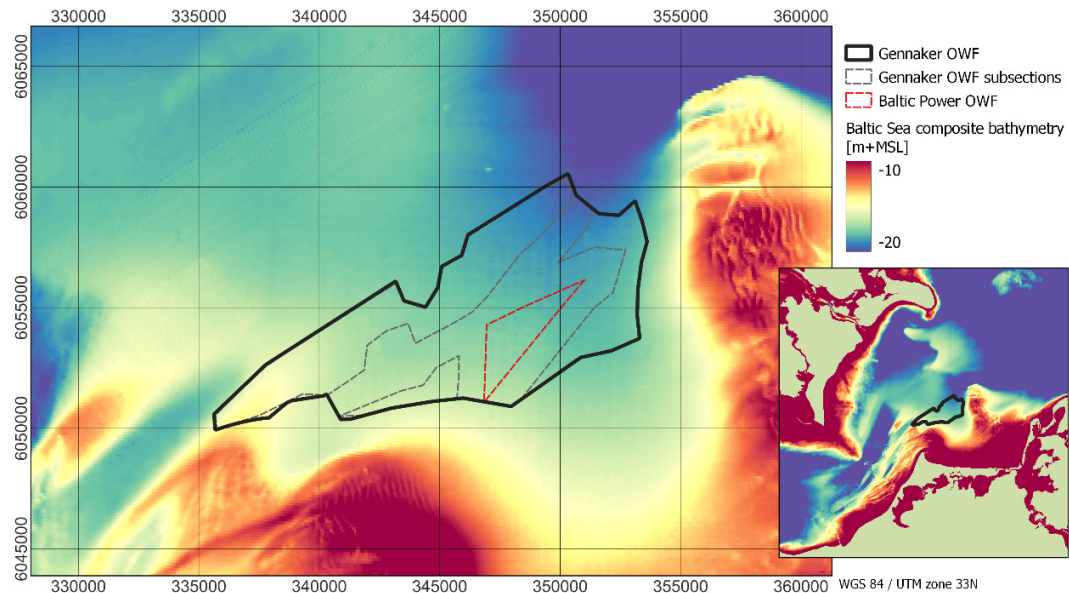


Figure 2.6 Baltic Sea composite bathymetry [m+MSL]. The lower right panel shows the Gennaker OWF on a larger scale.

2.4 Geotechnical and geophysical data

The sources for the geotechnical and geophysical data consists of (i) boreholes and cone penetration tests (CPTs) at five locations in the Gennaker OWF and (ii) a relatively coarse geological map of the Baltic Sea of the European Marine Observation and Data Network (EMODnet) (EMODnet Bathymetry Consortium, 2020). The CPTs and boreholes were performed by Fugro (Furgo, 2016). The boreholes allow classification of the sediments at different depths (lithology, basis sedimentary structures). The CPT logs provide geotechnical characteristics of the sediments to the penetration depth (e.g., tip resistance and sleeve friction values) allowing for the interpretation of the vertical sediment profile properties, mainly lithology and stiffness. The geological map of the Baltic Sea is used to get an impression of the seabed substrate for the wider area around the Gennaker OWF.

Figure 2.7 shows the locations of CPTs and boreholes located around the border of the Gennaker OWF. This figure also contains an overview of the seabed substrate for the wider area around the OWF. It is stressed that no metadata was available for the seabed substrate map, which is most likely based on a limited number of measurements.

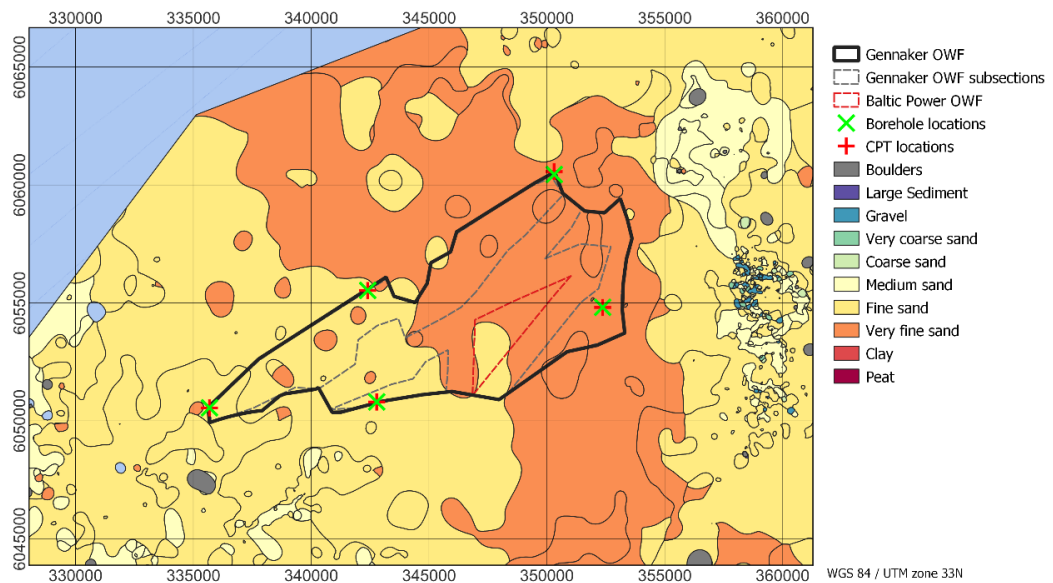


Figure 2.7 Borehole locations (Green cross), CPT locations (red plus) and an overview of the seabed substrate in and around the Gennaker OWF.

3 Seabed mobility study

3.1 Introduction

The objectives of the seabed mobility study are to assess the morphodynamics and provide an estimation of the future seabed levels in and around the Gennaker OWF. To achieve these objectives, an analysis is performed that consist of three parts: overview of the characteristics in the Gennaker OWF (Section 3.2), morphodynamic analysis including data-analysis and an extreme event assessment (Section 3.3) and estimation of future bed levels (Section 3.3.3).

3.2 Overview of the characteristics in the Gennaker OWF

The Gennaker OWF is located approximately 15 km north of the coast of Germany in the Baltic Sea. The area is characterised by mild tidal currents, a sloped seabed with local minor undulations and seabed substrate which predominantly consists of fine to very fine sand. This section gives an overview of the hydrodynamic, wave and the seabed geotechnical and geophysical characteristics in and around the Gennaker OWF.

3.2.1 Hydrodynamic and wave characteristics

A combination of several processes is driving the sediment dynamics at the Gennaker OWF. These governing processes include hydrodynamics (such as tides and wind-driven flow) and waves. Some of these processes are driven by atmospheric variations and meteorology (such as wind patterns, pressure fields, precipitation, etc.), which impose seasonal patterns, as well as episodic extreme events, on the system.

The Baltic Sea is a large enclosed shallow sea connected to the North Sea through the narrow and shallow Danish straits, with a very limited influence of the tide. Medvedev et al. (2016) describe that oceanic tides penetrate weakly or not at all in these types of enclosed basins (like the Black Sea and the Caspian Sea) but that tides are generated in the basins. The formation and predominance of diurnal or semi-diurnal tides depend on the frequency-selective properties of the basins. In the case of the Baltic Sea this means that in some regions diurnal tide dominates (e.g., in the Gulf of Finland), but most areas experience semi-diurnal tides. In the region of the Gennaker OWF, tidal amplitudes reach up to 0.15-0.20 m.

Figure 2.2 shows the roses of the depth-average total, tidal and residual current speed and direction for normal conditions at the three locations around the Gennaker OWF (see Figure 2.1). The roses show that currents range from 0.2 to 0.6 m/s, but are generally below 0.3 m/s. The most predominant depth-average total currents are oriented towards the east-northeast followed by west-southwest. Tidal current velocities are mild, with depth-averaged values mostly below 0.1 m/s. The residual currents generally make up the largest part of the total current velocity. For more information is referred to Medvedev et al. (2016) and Deltares (2022), but it can be concluded that the effect of the tide on the hydrodynamics in the Baltic Sea is mild.

Figure 2.3 shows the significant wave height and direction for the three locations around the Gennaker OWF. The significant wave height at the OWF is generally below 1.5 m (roughly 90% of the time) with associated peak wave periods below 7.0 s. The monthly roses (not included in this report) by Deltares (2022) show that the waves predominantly approach from the west-southwest (during the winter) or the northeast (during the summer). The most extreme waves generally come from the northeast. Moreover, the most severe and persistent normal conditions are found at the northern point of the Gennaker OWF (at location 101 in Figure 2.1).

3.2.2 Seabed characteristics

This section discusses the seabed characteristics around and in Gennaker OWF.

3.2.2.1 Regional scale

The bathymetry around the Gennaker OWF is shown in Figure 3.1. Water depths vary between 0 (at the coast) and >25 m (southwest and northeast of the Gennaker OWF). Close to the Gennaker OWF two areas with bedforms are observed. These areas are indicated with green circles and are mainly located in bed levels above -11 m relative to MSL. The bed features have heights in the order of one metre and lengths of approximately 250 to 400 m. The areas around the Gennaker OWF are further defined by flat seabeds with limited slopes or indications of bedforms. It is noted that the resolution of the Baltic Sea composite bathymetry is too limited to get indications of the presence of smaller scale bedforms such as megaripples.

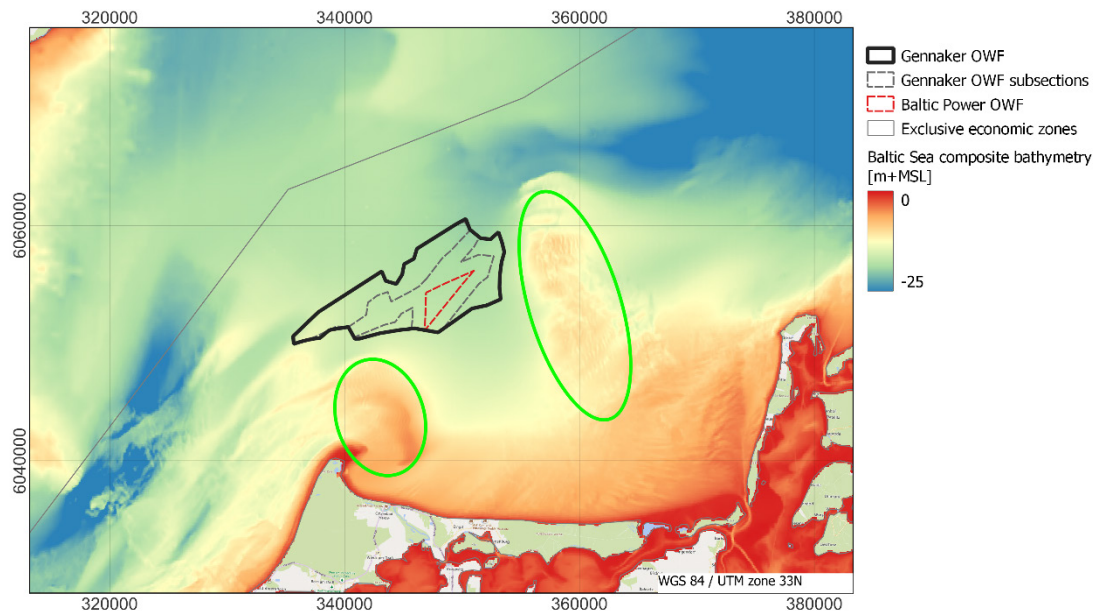


Figure 3.1 Overview of the bathymetry around the Gennaker OWF. Green circles indicate areas with bedforms.

3.2.2.2 Gennaker OWF

The seabed in the Gennaker OWF can be characterised as a sloped seabed with local minor undulations. An overview of the bathymetry in and around the Gennaker OWF is presented in Figure 3.2 (top panel). The bathymetry in this figure is created by overlaying the Gennaker 2022 bathymetry over the Gennaker 2013-2015 bathymetries and Baltic Sea composite bathymetry, respectively. Figure 3.2 (bottom panel) shows the bathymetry along the transect drawn from the southern border to the most northern point of the Gennaker OWF.

From the figures it can be seen that the bed level in the Gennaker OWF ranges from -14 to -20 m MSL. The water depth is shallowest at the south-western boundary of the OWF and increases steadily from southwest to northeast. Four areas in which notable features are observed in and around the Gennaker OWF are marked in Figure 3.2. These notable areas are analysed in more detail below.

Vermessungsbüro Weigt (2022) mentions that the 2013-2015 bathymetries have a positive offset of approximately 0.25 m with respect to the 2022 bathymetry. This is clearly visible in the bathymetries along Transect 1 (see Figure 3.2).

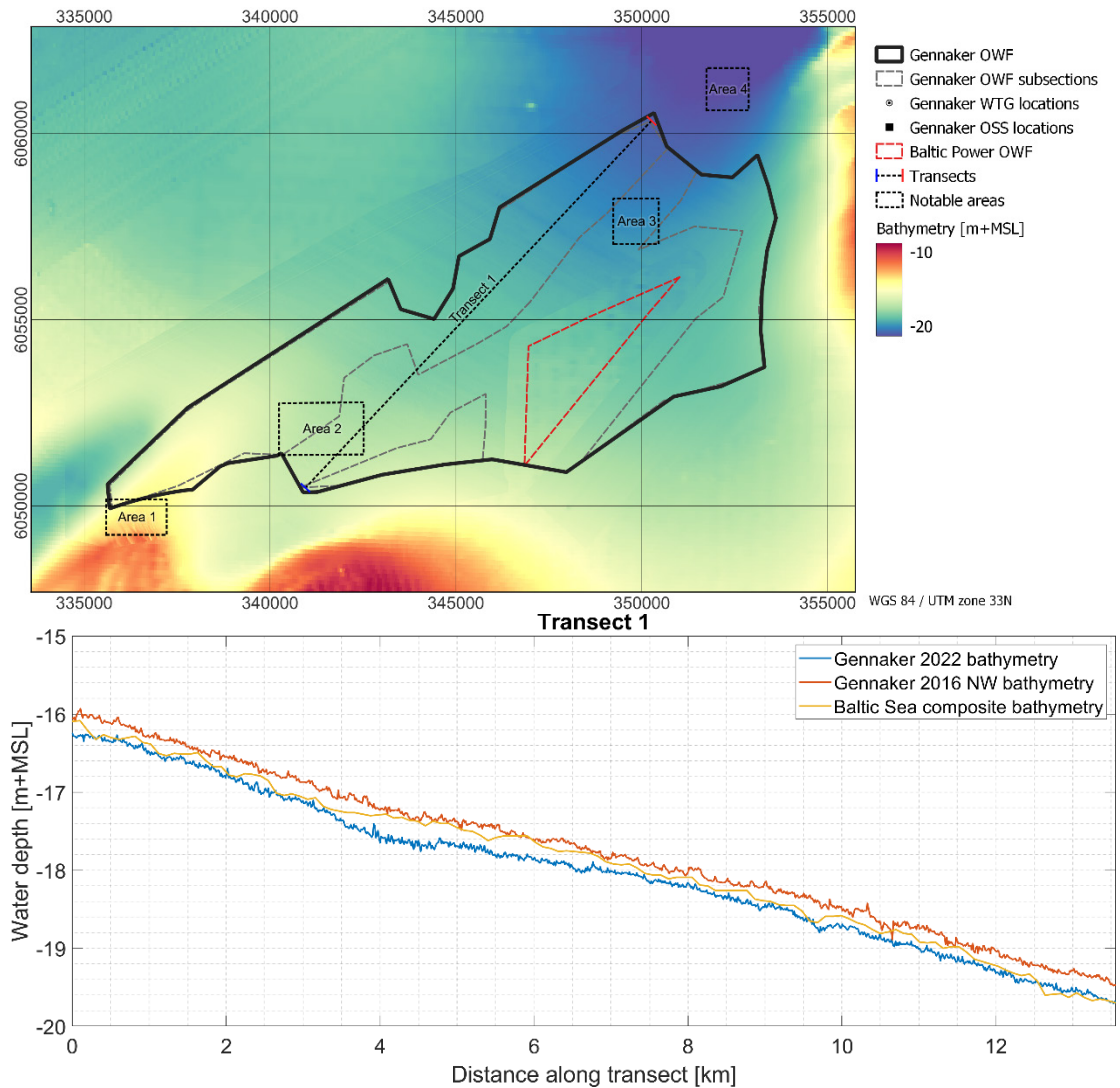


Figure 3.2 Overview of the bathymetry in and around the Gennaker OWF (top panel). The black dotted boxes indicate areas in which notable bed features are observed. Transect 1 is indicated by a black dotted line of which the start and end are marked by blue and red lines, respectively. Bathymetries along Transect 1 (bottom panel).

Figure 3.3 shows the megaripples that are observed just south of the most western point of the OWF area (see Area 1 in Figure 3.2). These bed features have amplitudes up to 0.25 m and lengths of several metres (see Transect 2). These megaripples are most likely also present in the areas marked in Figure 3.1 but cannot be observed due to the limited resolution of the Baltic Sea composite bathymetry.

The presented transect in the bottom panel of Figure 3.3 highlights the dynamics of the megaripples. It is observed that over the period 2015 to 2022 the megaripples change in dimensions and in position. Based on literature (Ashley, 1990) and morphological studies in the North Sea (e.g. Deltares, 2020b) megaripples are classified as highly dynamic with the potential to migrate over a full wavelength within a couple of months. The comparison of the 2015 and 2022 bathymetry data did not yield clear correlations between individual megaripples and therefore, no quantification of dynamics could be made. It is also observed that the averaged pattern, i.e. the seabed excluding the megaripples, shows a similar pattern in 2015 and 2022. For example, the troughs located at 0.25 and 0.45 km along the transect remain in

place. It is therefore concluded that in this specific area the seabed levels are as a minimum described by the upper and lower envelope of the megaripples. No significant changes were observed in megaripples dimensions. It is however stressed that this is based on only two high-resolution measurements. Furthermore, the megaripple dimensions are highly dependent on hydrodynamic and wave conditions and are subject to changes. Therefore, uncertainties are associated with the megaripple envelopes.

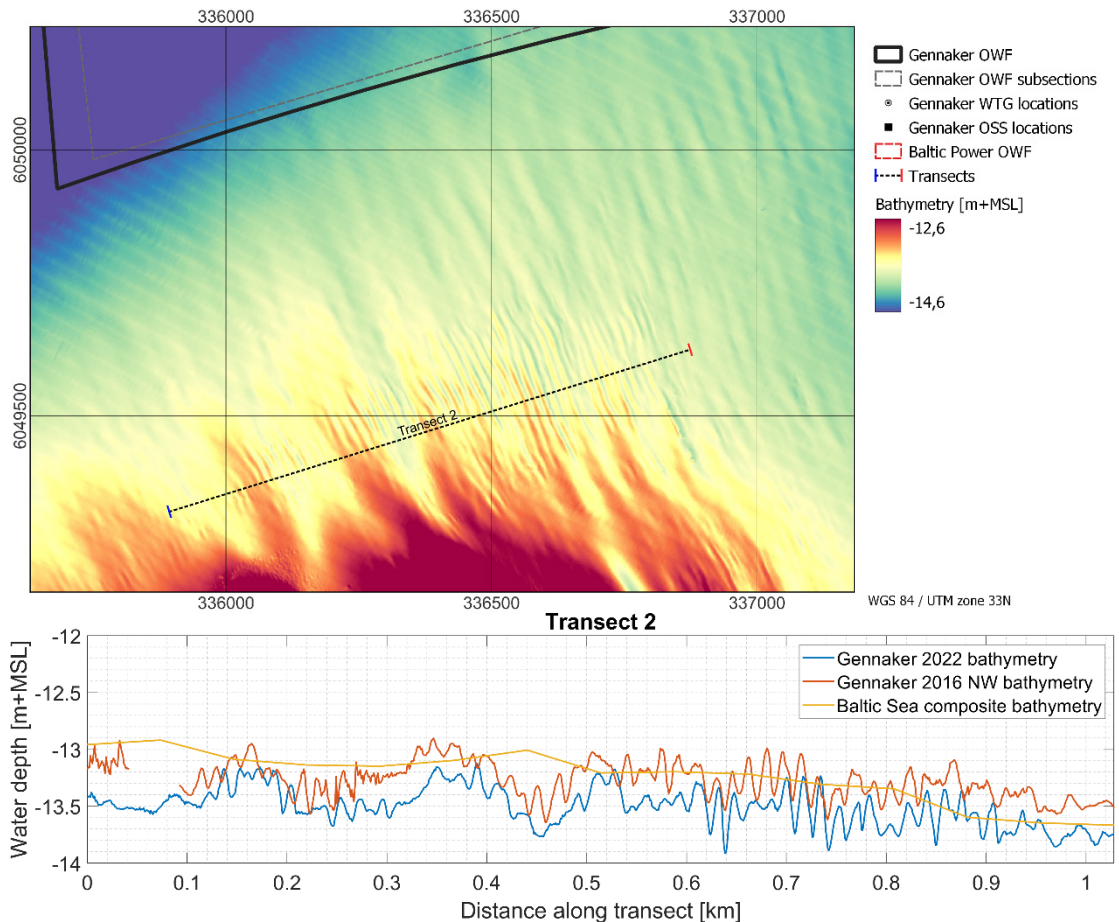


Figure 3.3 Megaripples located to the south of the most western point of the Gennaker OWF (see Area 1 in Figure 3.2) (top panel). Transect 2 is indicated by a black dotted line of which the start and end are marked by blue and red lines, respectively. Bathymetries along Transect 2 (bottom panel).

Small-scale features are observed throughout the Gennaker OWF, often occurring in clusters (distinct areas where multiple features are scattered around the area with a high density of occurrence), see for example Figure 3.4 (representing Area 2 in Figure 3.2). These features have a diameter of approximately 10 to 30 m and are typically about 0.30 m high with steep edge slopes (see Transect 3). In Vermessungsbüro Weigt (2022), these features are referred to as 'sediment structures', and suggested to be characterised by a surficial deposit with coarser sediment grain size compared the sediment adjacent and below the feature, based on prior SSS and SES surveys. It is highlighted that based on the interpretation of these features by Deltares (shape, slope, and surface characteristics), that these features may also consist of more erosion resistant cohesive muddy (i.e., finer) sediment, from which occasional edge erosion may occur. Such features are possibly deposited during the last glacial period. In this report, we continue to adopt the term 'sediment structures' while referring to these small-scale morphological features, consistent with the terminology used in Vermessungsbüro Weigt (2022).

Due to their limited erodibility (also able to sustain significant bed gradients along their edges), as well as mild near-bed hydrodynamics, it is anticipated that the sediment structures are immobile. This is also supported by their presence in the Gennaker 2015 bathymetry at identical locations (albeit with less pronounced signature, possibly due to data processing, as further discussed in Section 3.3.1), suggesting no migration of the features. Moreover, adjacent to some of these features, small but distinguishable scour pits are observed (typically directly east, which is in line with dominant near-bed flow directions, see Figure 2.2). Due to the mild near-bed hydrodynamics (Figure 2.2), the time required to develop such scour pits is anticipated to be very long, suggesting both a long presence of the features, as well as minor to no migration rates.

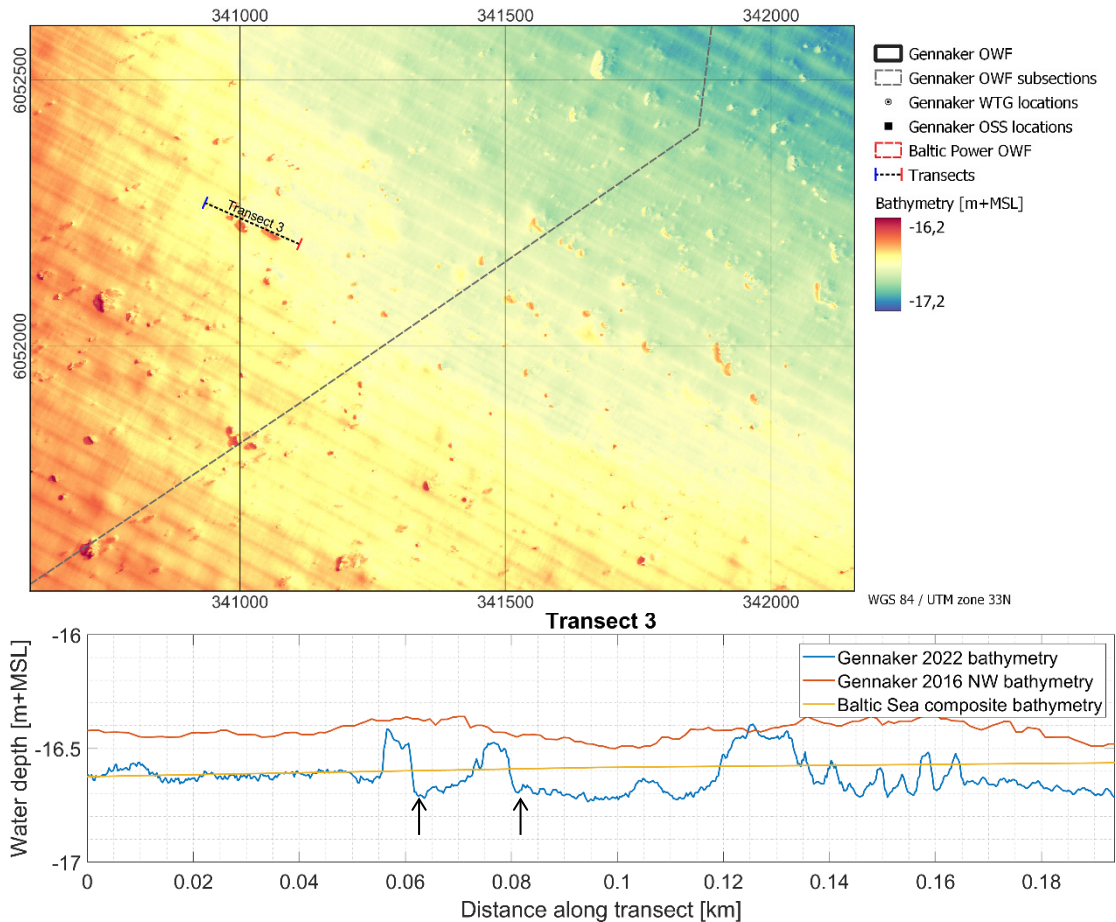


Figure 3.4 Example of ‘sediment structures’ located throughout the Gennaker OWF (Area 2 in Figure 3.2) (top panel). Transect 3 is indicated by a black dotted line of which the start and end are marked by blue and red lines, respectively. Bathymetries along Transect 3 (bottom panel). The small scour pits around these ‘sediment structures’ are indicated by black arrows.

Figure 3.5 shows two locations at which the seabed has been disturbed due to cable repairs along the high voltage cables that travel through the Gennaker OWF. These features span an area of approximately 300 and 450 m along the cable corridor and have a width smaller than 50 m (see Transect 4 and 5). The seabed disturbances are also present in the Gennaker 2015 bathymetry, in which case it is represented by trenches (likely prior to trench backfill, and possibly prior to cable installation).

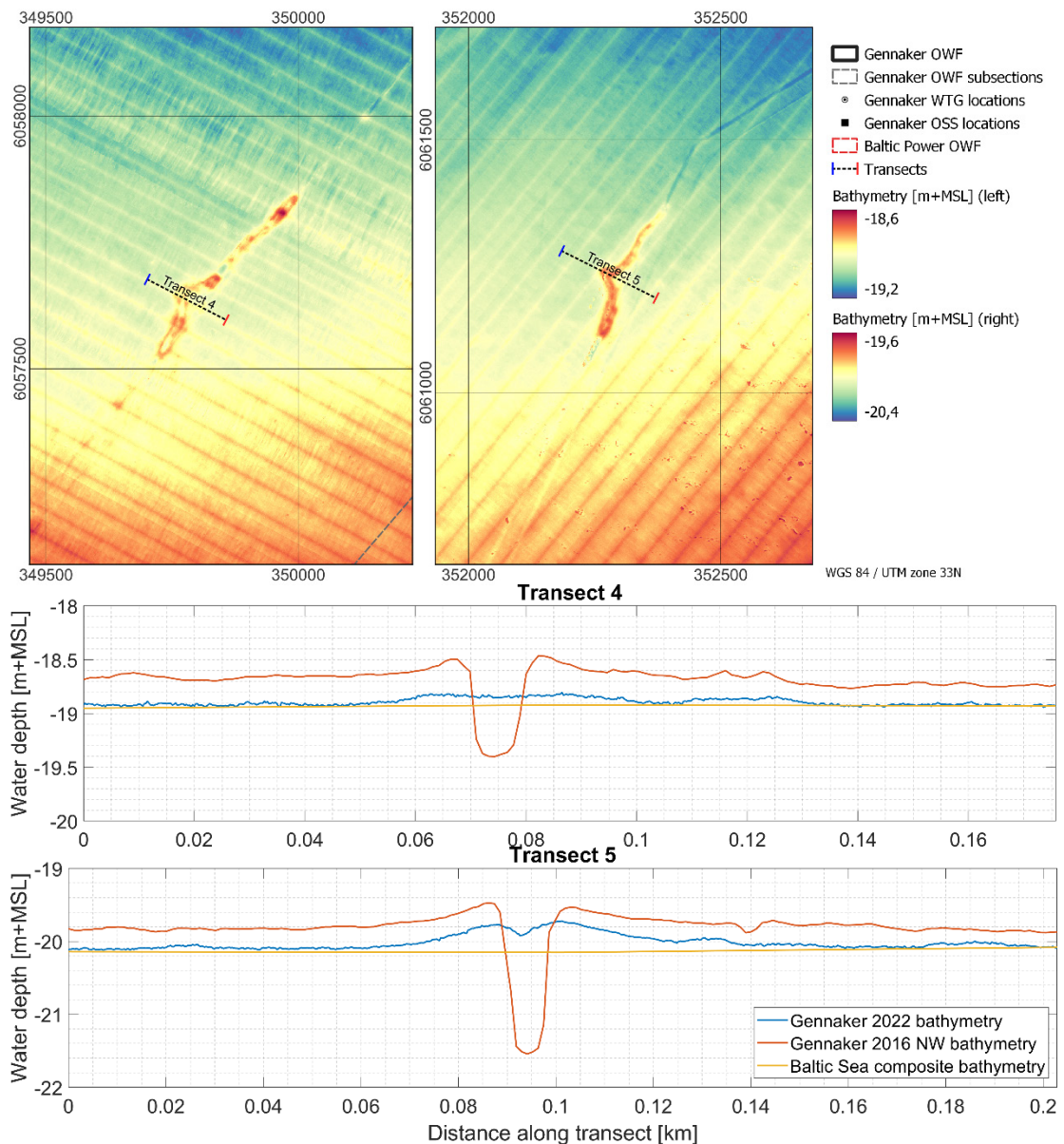


Figure 3.5 Cable repairs along the cable high voltage cables that travel through the Gennaker OWF (Area 3 and 4 in Figure 3.2) (top panel). Transect 4 and 5 are indicated by black dotted lines of which the start and end are marked by blue and red lines, respectively. Bathymetries along Transect 4 and 5 (centre and bottom panel).

3.2.3 Geotechnical and geophysical characteristics

Sediment dynamics are governed by the type of sediment available in the seabed. This section focuses on the present sediment characteristics and geology of the Gennaker OWF and its surroundings.

The present-day Baltic Sea exists in its current shape for only the last few thousand years. The Baltic Sea is considered to have a tectonic origin with further erosion during glacial periods in the Pleistocene era. The repeated cycles of glacial and interglacial periods resulted in eroded bedrock overlain by a variable thickness of mixed glacial and glacio-lacustrine deposits (this includes coarse (boulders, cobbles and gravel) and finer material (sand and over-consolidated silts and clays)) (ODE Group Ltd, 2021). This is further discussed in Voipio (1981), Zachowicz et al. (2004) and Kaskela et al. (2012).

Sediment accumulation since the last ice age is considered to be from marine sedimentation and from the erosion and reworking of glacial and glaciolacustrine deposits (ODE Group Ltd, 2021). Vallius et al. (2020) gives an overview of the quaternary geology (i.e., the origin of the surficial sediments). Based on this overview, the Gennaker OWF has Holocene sediment on the present seabed.

An important contributor to sediment transport and seabed morphodynamics is the type of sediment present at the seabed. An overview of the seabed substrate in and around the Gennaker OWF is given in Figure 2.7. This figure shows that the sediment in the OWF is predominantly classified by very fine sand in the northeast and fine sand in the southwest. The five borehole and CPT logs comply with these observations. These show that the top layer consists mainly of fine sand and has depth ranging from 9 m at the to 25 m. Figure 2.7 shows that the substrate in the area surrounding the Gennaker OWF also consist of very fine to sand. However, locally and to the east of the OWF medium to very coarse sand and gravel is present.

3.3 Morphodynamic analysis

This section presents the morphodynamic analysis in and around the Gennaker OWF. First a data-driven analysis is presented focussing on seabed level changes in a regional setting (around Gennaker OWF) and in a local setting (at Gennaker OWF). This is followed by a numerical simulation of an extreme event focussing on the impact on seabed levels

3.3.1 Regional analysis

Within the Gennaker OWF limited seabed changes are observed as discussed in previous sections. This section provides an insight in the more regional scale seabed level variations.

As discussed in Section 2.3, limited data is available outside the Gennaker OWF to a single composite bathymetry for the Baltic Sea derived from Bundesamt für Seeschifffahrt und Hydrographie (BSH) (2022) and EMODnet Bathymetry Consortium (2020). The bathymetry data available for the German part of the Baltic Sea is interpolated from multiple measurements conducted over the period 1992 to 2016. The available data however provides, next to the interpolated bathymetry, information on the minimum and maximum measured seabed levels. Figure 3.6 shows the difference between the minimum and maximum measured seabed levels. The figure indicates significant areas with limited (<0.25 m) differences, among which the Gennaker OWF, and areas east and west of Gennaker with more pronounced differences. It is noted that this difference is often between two measurements with different resolutions. Or just based on a single measurement. Absence of differences might therefore not directly indicate limited seabed level changes.

Considering the areas just outside Gennaker OWF, three observations are made. First, the areas highlighted with green circles indicate migration of bed forms. This is however limited to areas shallower than 11 m MSL (see Section 3.2.2). No indications were found of bedform presence in the Gennaker OWF. The second area is indicated by a red circle. Within this area numerous significant seabed variations are observed.

These observed differences in the red circled area can most likely be attributed to data resolution. Figure 3.7 shows the sediment types in the German part of the Baltic Sea. Comparison with Figure 3.6 learns that in these areas both boulders and larger sediment sizes were found. Often these areas are characterised by a more rugged terrain. Differences in data resolution, especially with the coarser resolution available for the Baltic Sea composite bathymetry, can results in significant differences between measurements.

Lastly, just west of the Gennaker OWF a couple of southwest to northeast lines with differences are found. These are also most likely a result of data resolution as they coincide with slopes in the seabed.

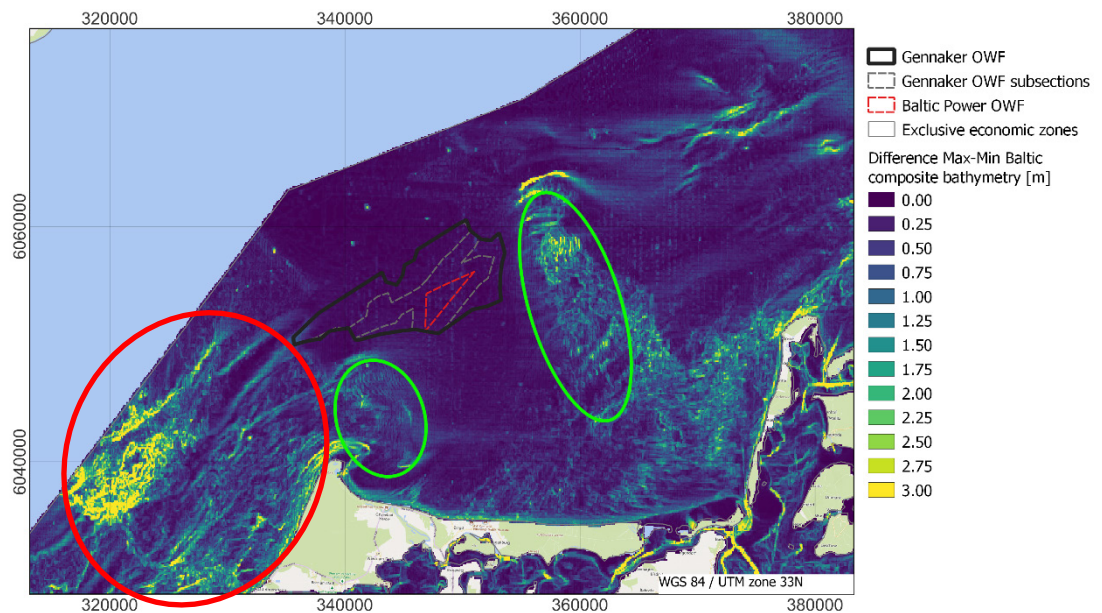


Figure 3.6 Overview of the differences in water depths levels around the Gennaker OWF based on the max and min soundings of the Baltic composite bathymetry. Green circles indicate areas with bedforms (see also Section 3.2.2.1). The red circle indicates areas with significant differences.

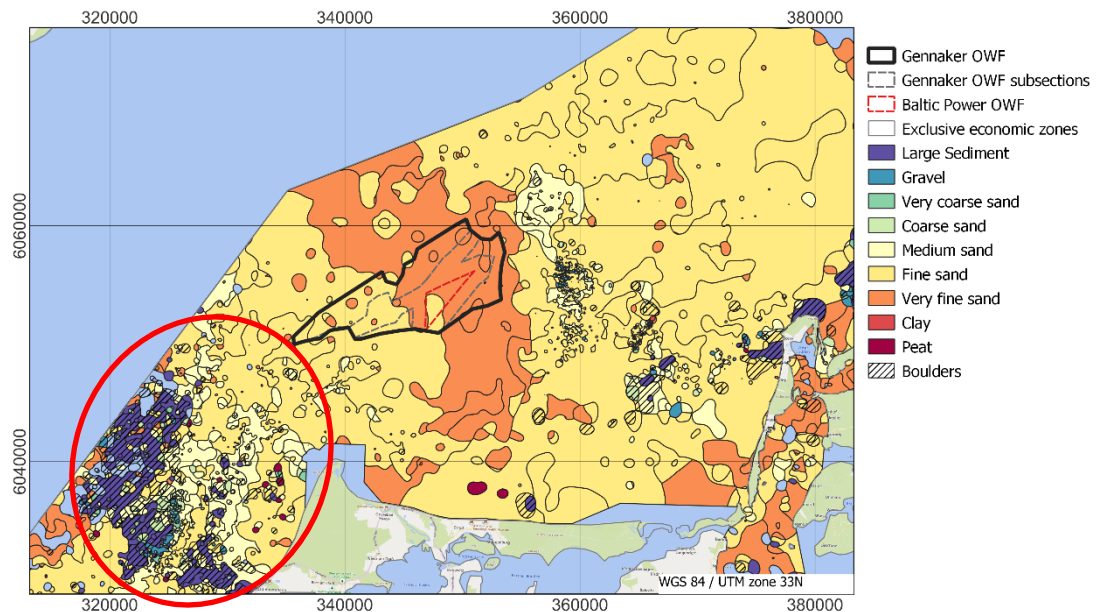


Figure 3.7 Overview of the seabed substrate in the Baltic Sea. The red circle indicates areas with significant differences as shown in Figure 3.6.

3.3.2

Seabed level changes

To create an understanding of seabed level changes in and around the Gennaker OWF comparisons of the available bathymetric data are made. Multiple bathymetry datasets are available, allowing us to determine if bathymetry changes have occurred over time. This analysis will also demonstrate to what extent the seabed is mobile. It is important to make this characterisation and to determine the extent of the seabed mobility since the seabed can vary under different conditions.

Figure 3.8, Figure 3.9 and Figure 3.10 present an overview of differences in seabed levels in the Gennaker OWF based on comparison of the available bathymetry datasets presented in Table 2.1. All comparisons are made with respect to the most recent bathymetry (i.e., the Gennaker 2022 bathymetry). No corrections have been applied on the datasets. The changes that can be observed in these figures might not be related to seabed dynamics but might just be the result of uncertainties and differences in the vertical reference levels of the multibeam surveys.

Figure 3.11 contains three histograms that show the distribution of seabed level differences for the bathymetry comparisons presented in Figure 3.8, Figure 3.9 and Figure 3.10. This figure also shows the mean and standard deviation for the seabed level differences.

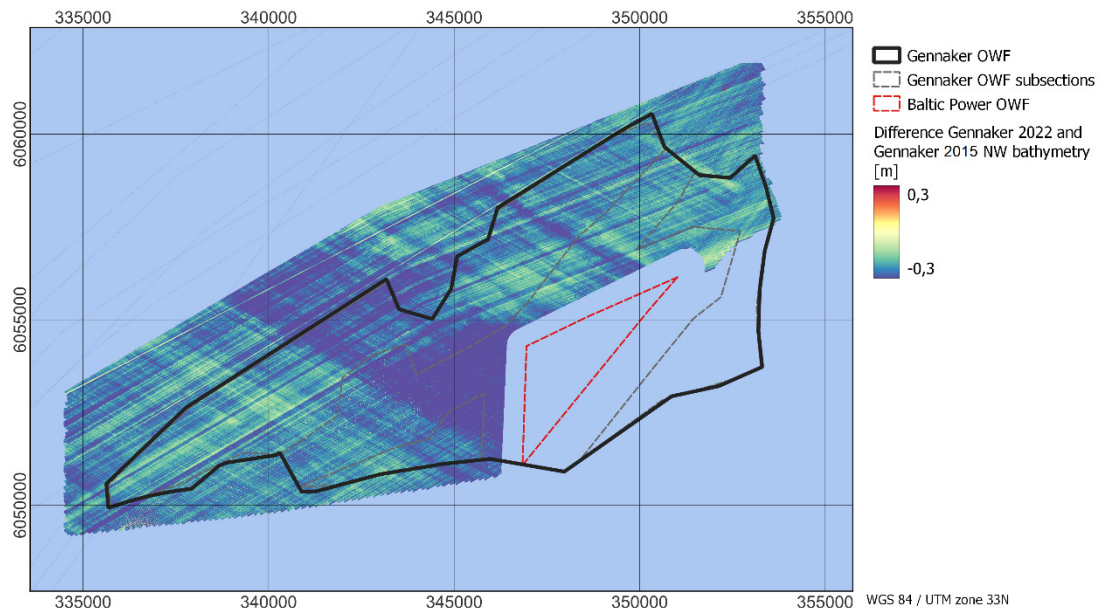


Figure 3.8 Difference between Gennaker 2022 and Gennaker 2015 NW bathymetry [m].

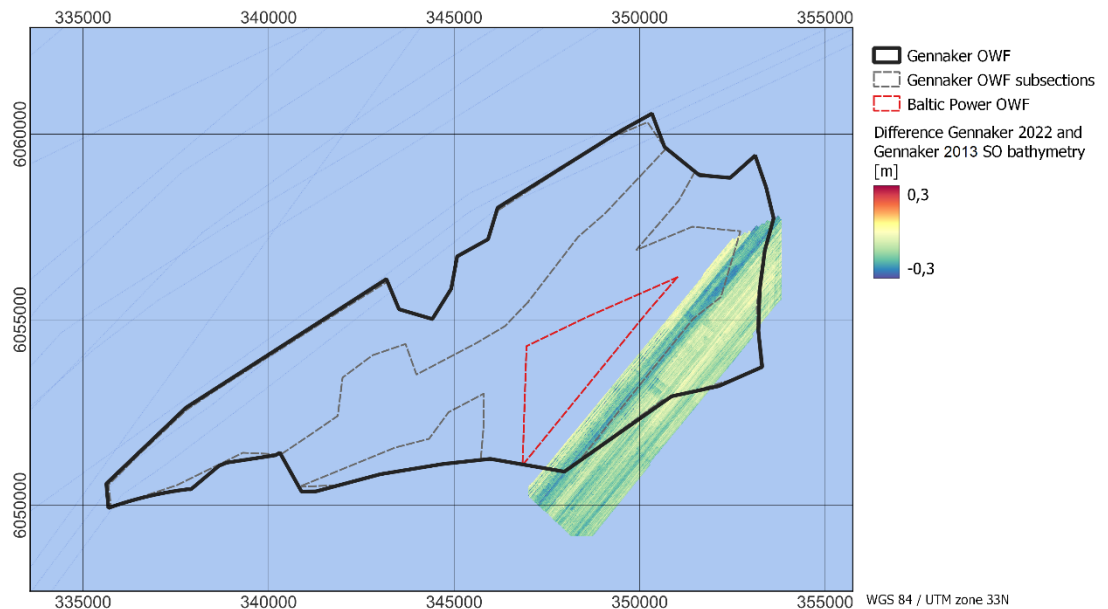


Figure 3.9 Difference between Gennaker 2022 and Gennaker 2013 SO bathymetry [m].

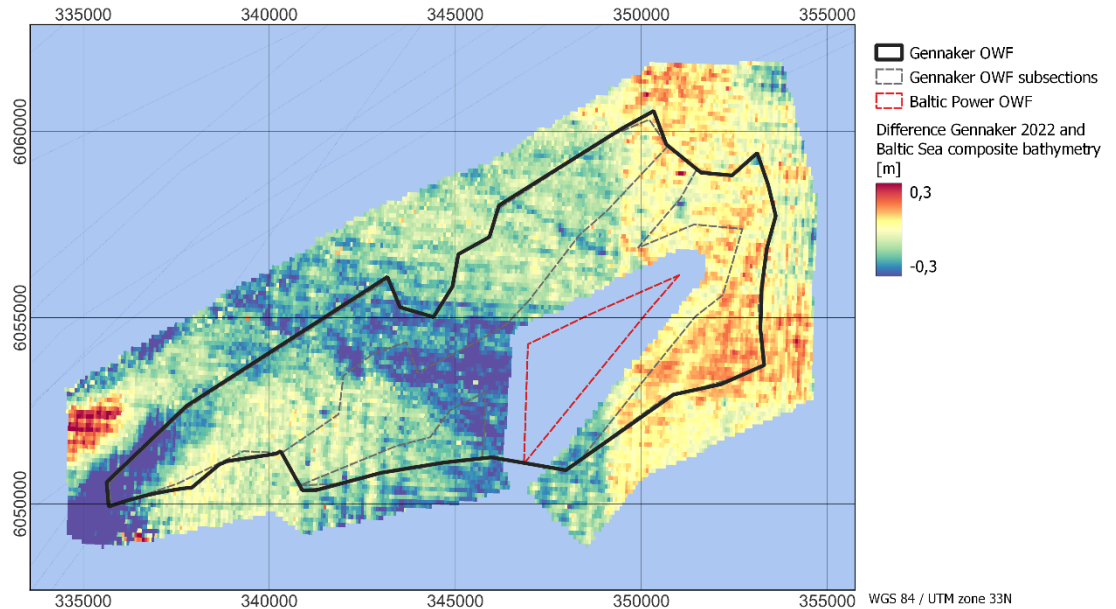


Figure 3.10 Difference between Gennaker 2022 and Baltic Sea composite bathymetry [m].

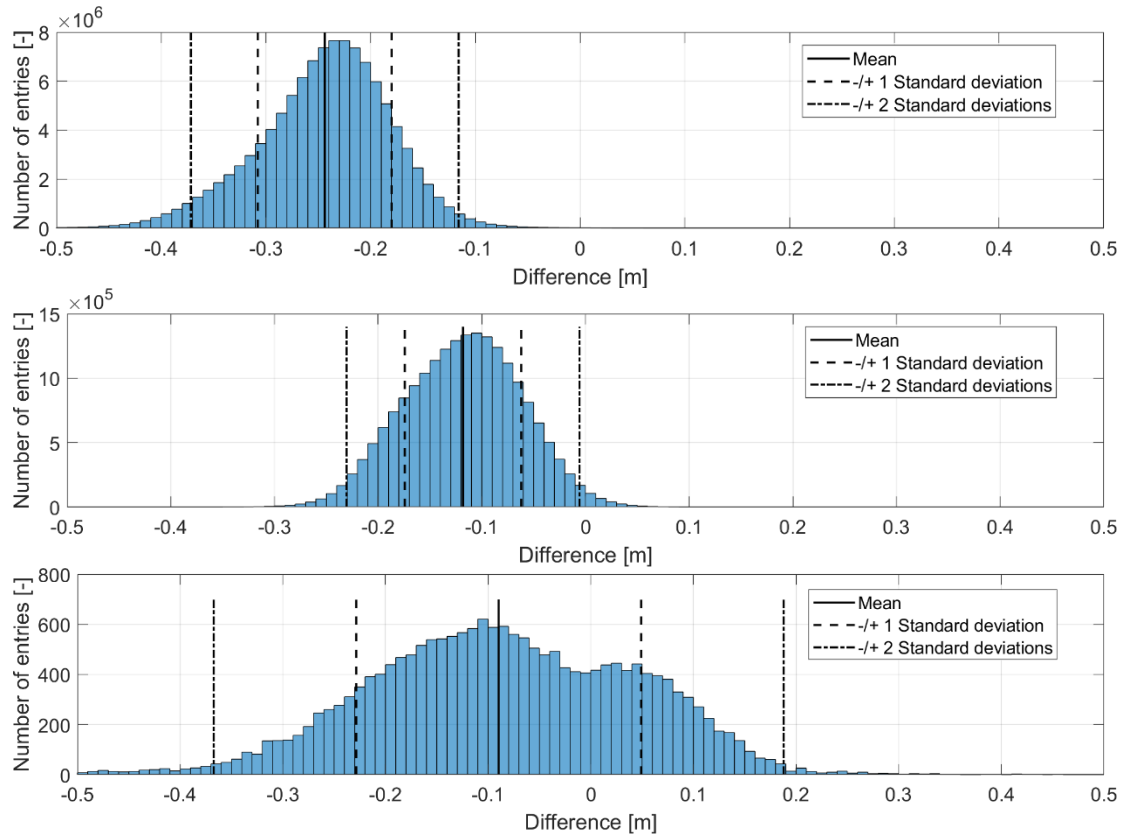


Figure 3.11 Differences between bathymetries presented in Table 2.1: difference between Gennaker 2022 and Gennaker 2015 NW bathymetry (top panel); difference between Gennaker 2022 and Gennaker 2013 SO bathymetry (centre panel); difference between Gennaker 2022 bathymetry and Baltic Sea composite bathymetry (bottom panel).

From the overview figures (see Figure 3.8 to Figure 3.10), the histograms (see Figure 3.11) and transects (see Figure 3.2 bottom panel) it can be observed that there are relatively constant offsets between the Gennaker 2022 bathymetry and the other three bathymetries. These

offsets are most likely caused by differences in survey equipment, data processing methods and referencing techniques used when creating the different datasets. Moreover, data artefacts at the interface between track lines of the multibeam surveys are also clearly visible in Figure 3.8 and Figure 3.9.

All three bathymetries are generally shallower than the 2022 bathymetry. The histograms show that the differences between the 2022 and 2013-2015 bathymetries clearly follow normal distributions. The histogram for the difference between the Gennaker 2022 and Baltic Sea composite bathymetry is less normally distributed. This is mostly likely because of the larger regional differences caused by the lower resolution of the Baltic Sea bathymetry (see Figure 3.10).

To get insight in bed level changes related to morphodynamic activities in the area corrections are made to the datasets. The aim of these corrections is to discover whether apart from the uncertainties in bed levels measurements and differences in vertical reference levels any other local bed level changes have occurred in the area. A spatially constant correction, equal to the mean of the differences (see Figure 3.11) is applied to each bathymetry. After correction, the mean of the differences for each bathymetry comparison is equal to zero. This means that the Gennaker 2015 NW, Gennaker 2013 SO and Baltic Sea composite bathymetry are decreased by 0.24 m, 0.12 m and 0.09 m, respectively.

The overviews of the differences in seabed levels in the Gennaker OWF after correction are presented in Figure 3.12, Figure 3.13 and Figure 3.14. The histograms of the corrected differences are shown in Figure 3.15. The histograms for the difference between the 2022 and corrected 2013-2015 bathymetries show that the majority of the observed differences (within two standard deviations) fall between -0.15 and 0.15 m. The differences between the 2022 bathymetry and Baltic Sea data are more widespread but generally (within two standard deviations) fall between -0.30 and 0.30 m. As mentioned before, this is mostly likely because of the larger regional differences caused by the lower resolution of the Baltic Sea bathymetry. The observed differences are very small compared to differences observed in areas with significant morphodynamic activity.

The limited seabed dynamics in the area are further highlighted when focusing on the notable areas in and around the Gennaker OWF (see black dotted boxes in Figure 3.2). The seabed level differences between the 2022 and 2015 bathymetry in these notable areas are shown in Figure 3.16 to Figure 3.18. These figures show that the largest changes in seabed levels due to sediment transport predominantly occur around the megaripples (see Figure 3.16) and is also suggested around the sediment structures (see Figure 3.17).

Figure 3.3 and Section 3.2.2.2 showed and discussed that over the period 2015 to 2022 seabed level variations in the area where megaripples are found is described by the envelope of the megaripples. Figure 3.16 confirms this conclusion. The figure shows the differences in bed levels along a transect in the megaripple area. After correction of the 2015 bathymetry data, it can be seen that differences vary with ± 0.40 m around a zero mean.

For the latter, it is however stressed that no clear migratory behaviour is observed, with distinct erosion and deposition patterns (as observed for the megaripples in Figure 3.16), as mostly only deposition of the sediment structures is suggested by the difference plot. This is not very likely (and not 'mass conserving'), which is also supported by Vermessungsbüro Weigt (2022), which suggests that these features were already present in 2015, based on SSS and SES surveys. As such, data inaccuracies due to post-processing (e.g., data averaging, smoothing) may have resulted in a misrepresentation of the small-scale sediment structures in the 2015 bathymetry. In Section 3.2.2.2, it is already anticipated that these sediment structures are mostly immobile. Regardless, accounting for the uncertainties in potential seabed level

changes around sediment structures is considered a conservative requirement in this study. Seabed level changes (i.e., approximately 0.30 m) are also observed along and cable repairs (see Figure 3.18). Finally, in all figures the data artefacts at the interface between track lines of the multibeam surveys are clearly visible.

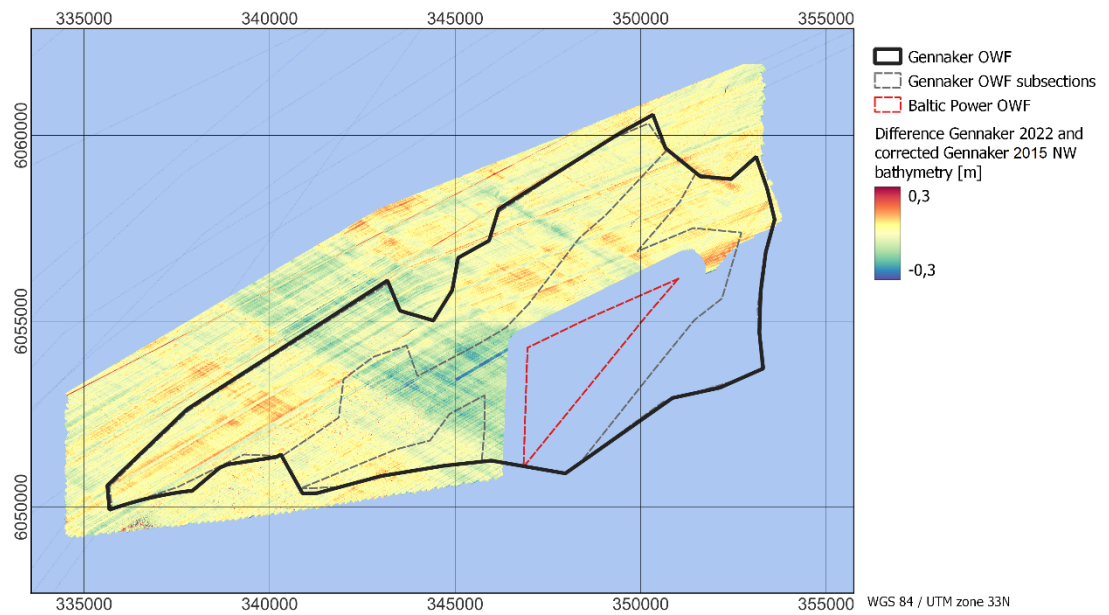


Figure 3.12 Difference between Gennaker 2022 and corrected Gennaker 2015 NW bathymetry [m].

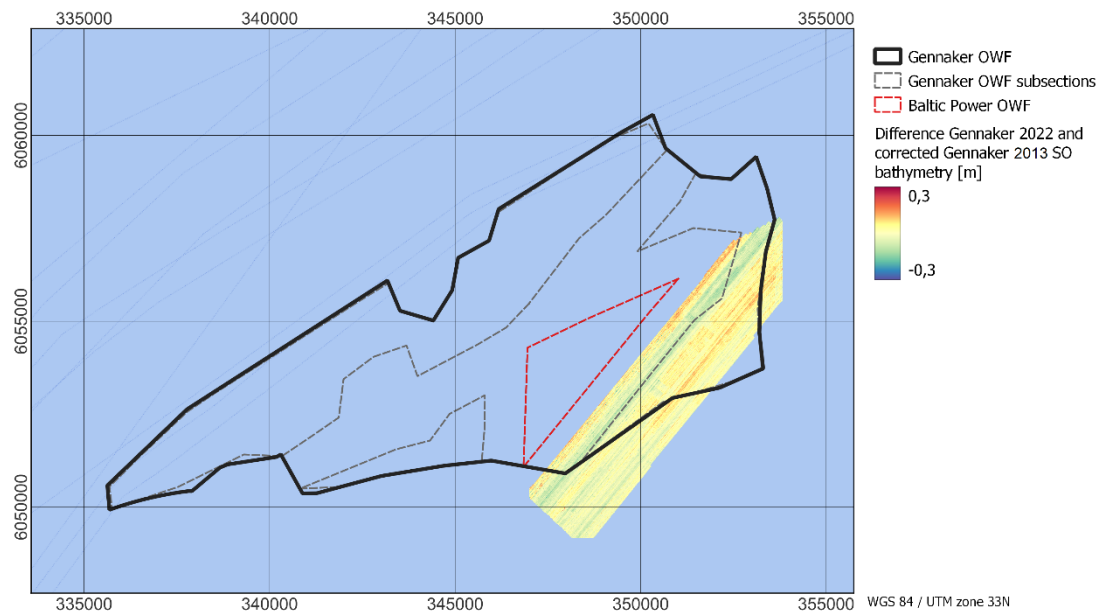


Figure 3.13 Difference between Gennaker 2022 and corrected Gennaker 2013 SO bathymetry [m].

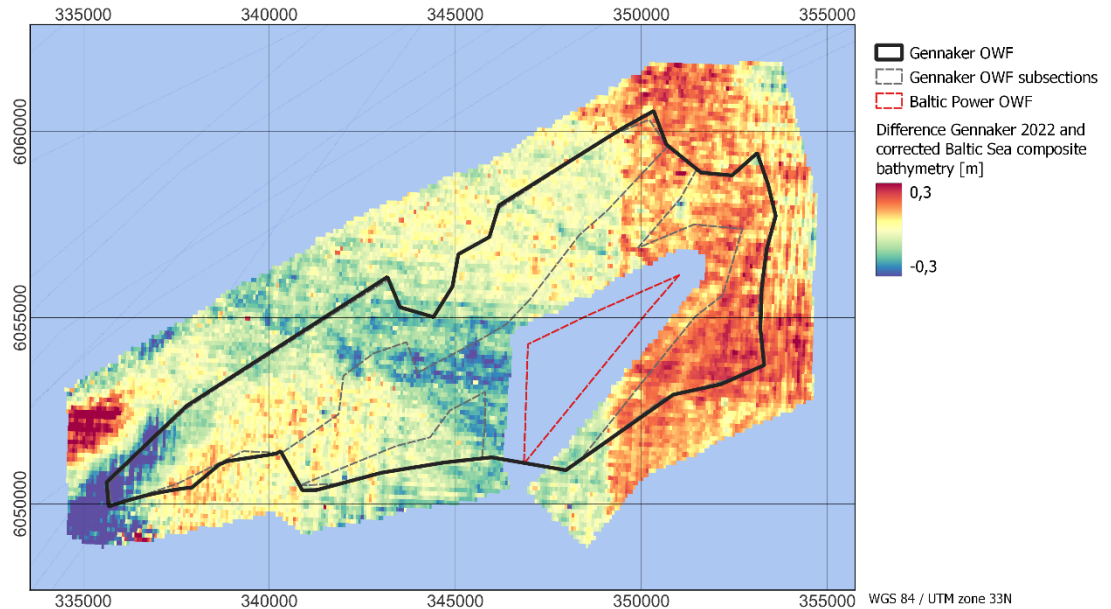


Figure 3.14 Difference between Gennaker 2022 and corrected Baltic Sea composite bathymetry [m].

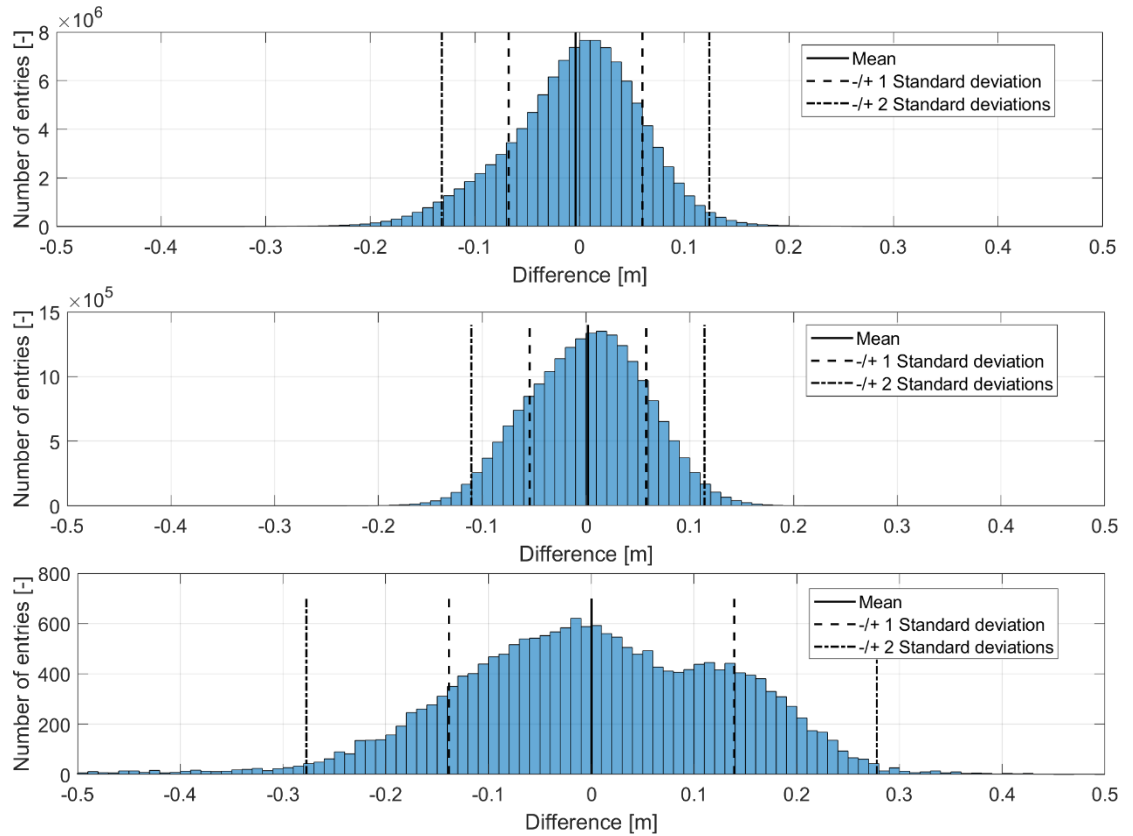


Figure 3.15 Differences between bathymetries presented in Table 2.1 after correction: difference between Gennaker 2022 and corrected Gennaker 2015 NW bathymetry (top panel); difference between Gennaker 2022 and corrected Gennaker 2013 SO bathymetry (centre panel); difference between Gennaker 2022 bathymetry and corrected Baltic Sea composite bathymetry (bottom panel).

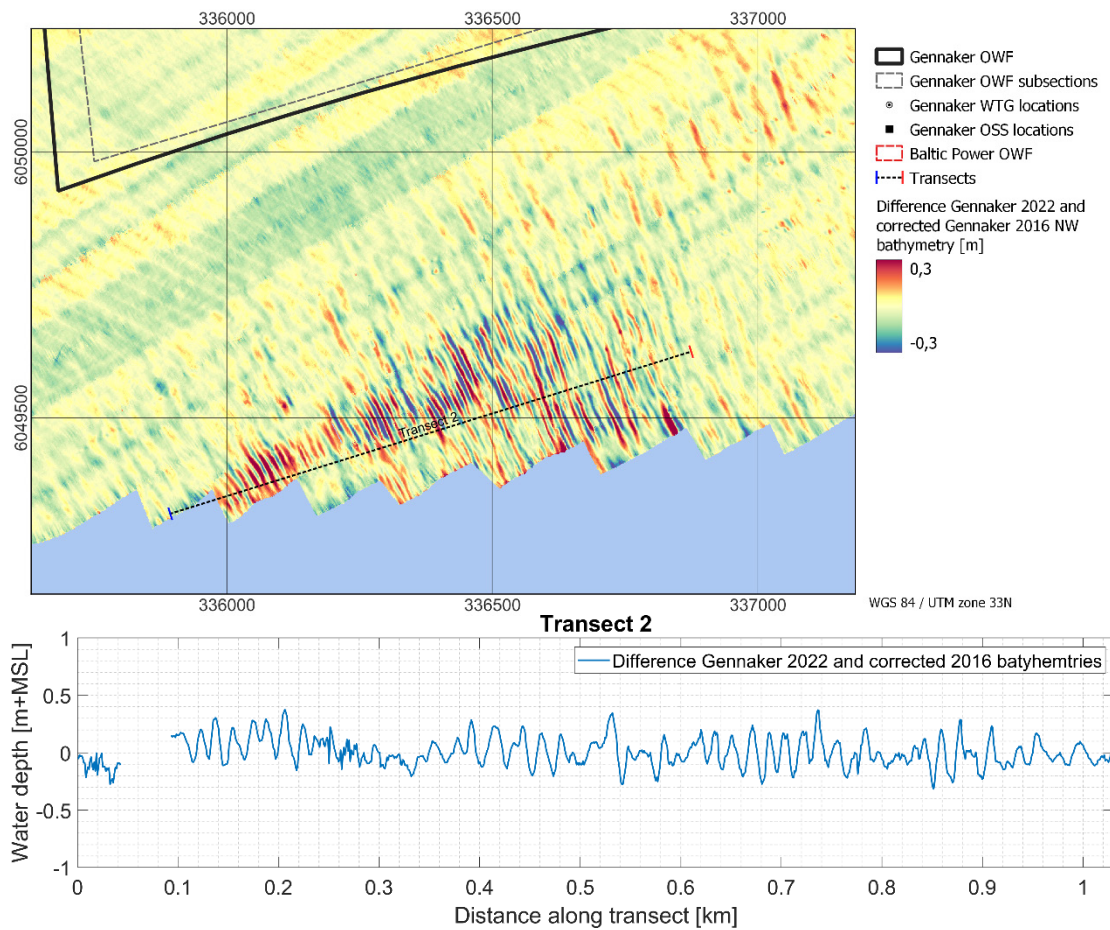


Figure 3.16 Differences between Gennaker 2022 and corrected Gennaker 2015 bathymetry for the megaripples located to the south of the most western point of the Gennaker OWF (see Area 1 in Figure 3.2) top panel). Transect 2 is indicated by a black dotted line of which the start and end are marked by blue and red lines, respectively. Bathymetry difference along Transect 2 (bottom panel).

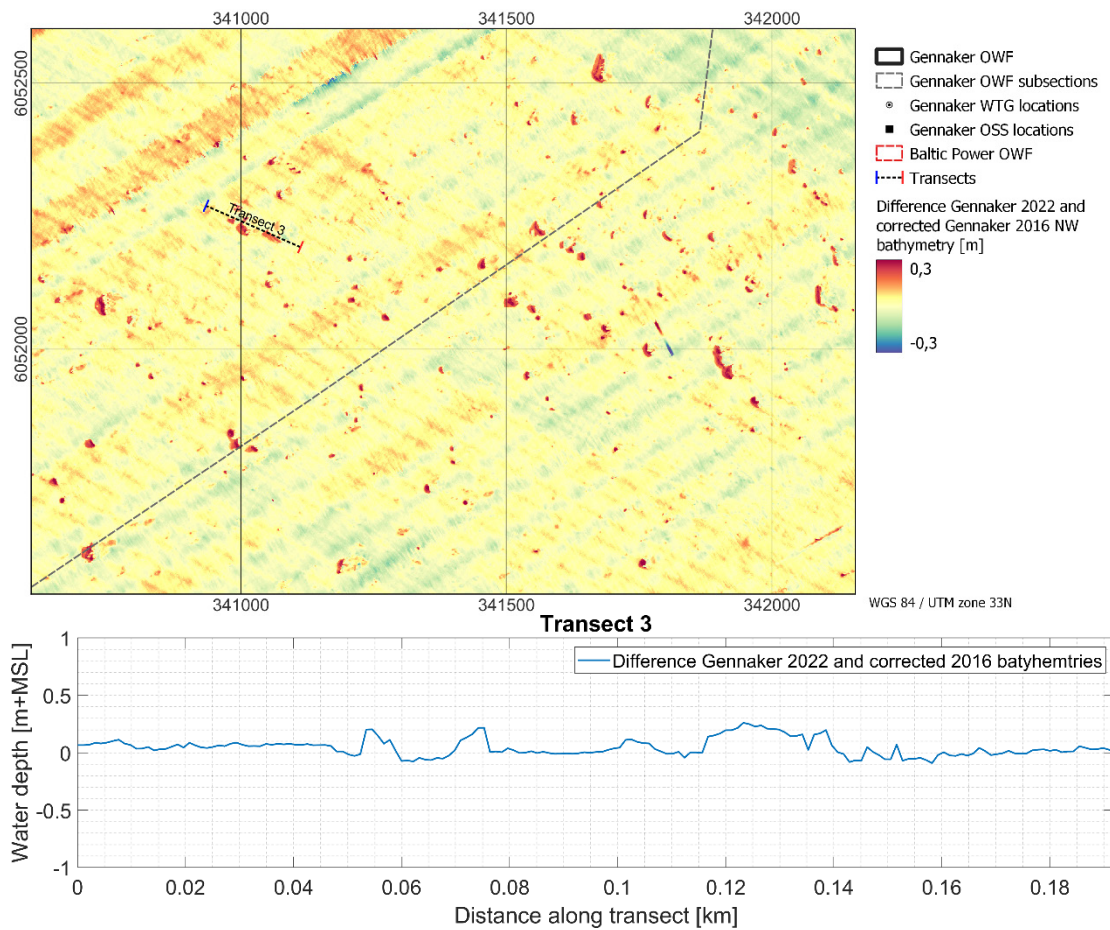


Figure 3.17 Differences between Gennaker 2022 and corrected Gennaker 2015 bathymetry for an example of sediment structures located throughout the Gennaker OWF (Area 2 in Figure 3.2) (top panel). Transect 3 is indicated by black dotted line of which the start and end are marked by blue and red lines, respectively. Bathymetry difference along Transect 3 (bottom panel).

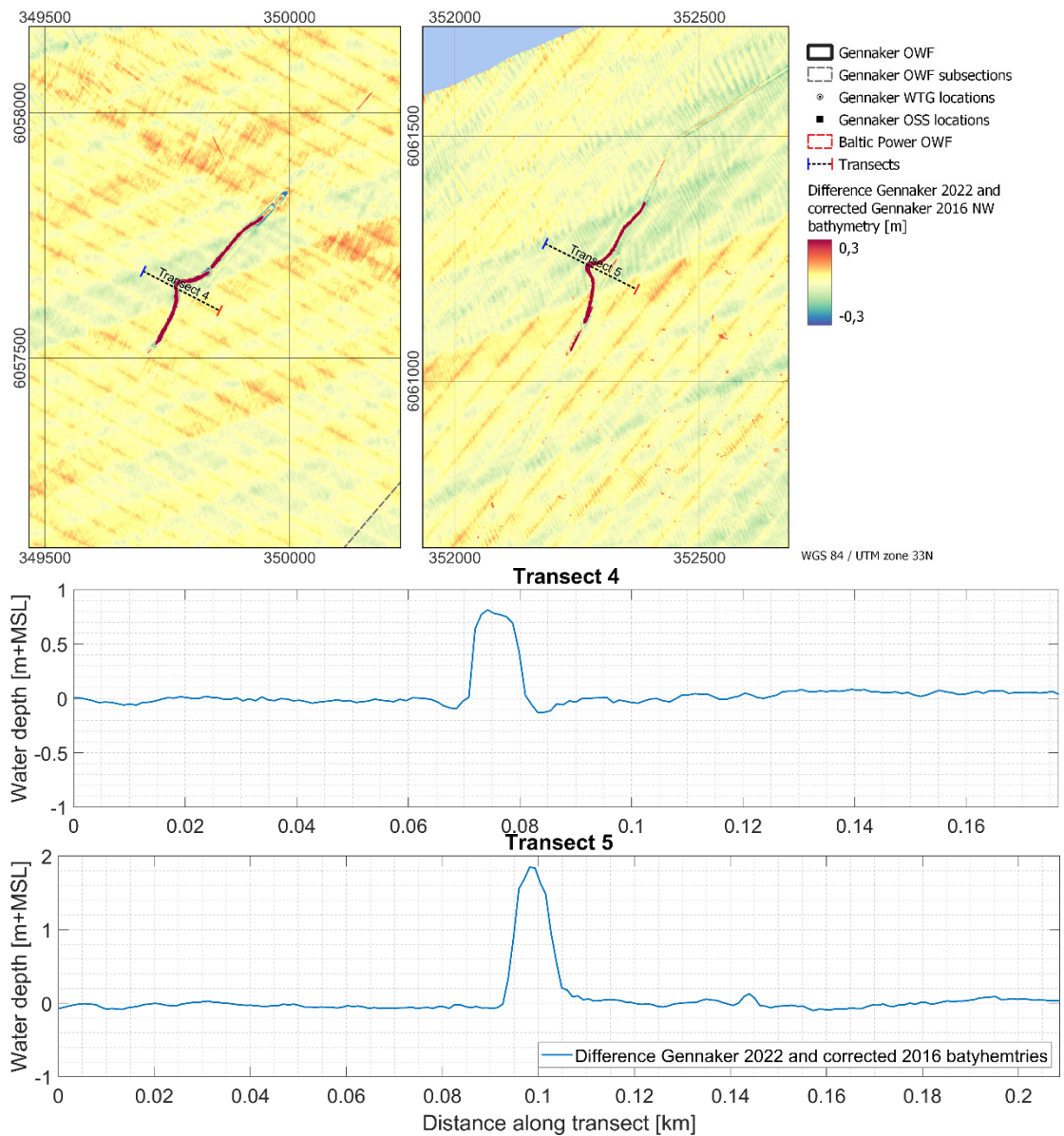


Figure 3.18 Differences between Gennaker 2022 and corrected Gennaker 2016 bathymetry for the cable repairs along the cable high voltage cables that travel through the Gennaker OWF (Area 3 and 4 in Figure 3.2) (top panel). Transect 4 and 5 are indicated by black dotted lines of which the start and end are marked by blue and red lines, respectively. Bathymetry difference along Transect 4 and 5 (centre and bottom panel).

3.3.3 Extreme event assessment

Sections 3.3.1 and 3.3.2 indicate limited mobility over the windfarm area. The low current velocities and wave heights present over the large majority of the year, described in Deltares (2022) and summarised in Section 2.2 further support this observation. Nevertheless, during extreme events the hydrodynamic conditions may become stronger, inducing larger and potentially significant bed level changes. In the absence of pre- and post-extreme event survey data available, numerical modelling was used to assess the impact of extreme events on the large-scale seabed and inform the seabed level predictions.

To assess the impact of extreme events on the large-scale seabed (i.e. not on a megaripple and sand wave scale) and support the data driven-driven analysis as there is no pre- and post-extreme event survey data available, a numerical model was setup in Delft3D-Flexible Mesh (FM) coupled with D-Wave, while including the sediment transport module. The model is used to simulate hydrodynamics, waves and sediment transports over Gennaker OWF during an

extreme event. The selection of a representative extreme event to be modelled is presented in Section 3.3.3.1. Section 3.3.3.2 describes the setup of the coupled hydrodynamic, wave and sediment transport model. Lastly, the results of the simulation are presented and discussed in Section 3.3.3.3.

3.3.3.1 Event selection

To select the representative storm conditions for Gennaker OWF an analysis that combines information and datasets from the metocean assessment for Gennaker OWF (Deltares, 2022) is made. More specifically, hourly wave timeseries produced by the Deltares' in-house SWAN hindcast model were extracted for one location in Gennaker OWF (at the southwest part of the Gennaker OWF, see the red dot in Figure 3.21). The available timeseries, presented in Figure 3.19, cover a period of more than 40 years from 1979 to 2021.

The storm that peaked on 03-Dec-1999 (marked with a green dashed line in Figure 3.19) is selected as the event with the maximum significant wave height at the reference point during the full hindcast period. This storm peak corresponds to [storm Anatol](#), that passed over the Danish and German North Sea moving in north-eastward direction over the Baltic sea. The evolution of the wind speed, computed current velocities and waves during the storm over the area of interest, are extracted for the reference location from the metocean modelled timeseries, is presented in Figure 3.20. Wind speeds reach 30 m/s, while (near-bottom) current velocity magnitudes reach 0.68 m/s at the reference location. The simulated significant wave height reaches 4.1 m at the reference location during the peak of the storm, by 20:00h of December 3, 1999. Compared to the metocean conditions for the nearby point 028 (Deltares, 2022) the presented wave conditions during storm Anatol (as modelled using the calibrated metocean model) correspond to a return period of 50 years.

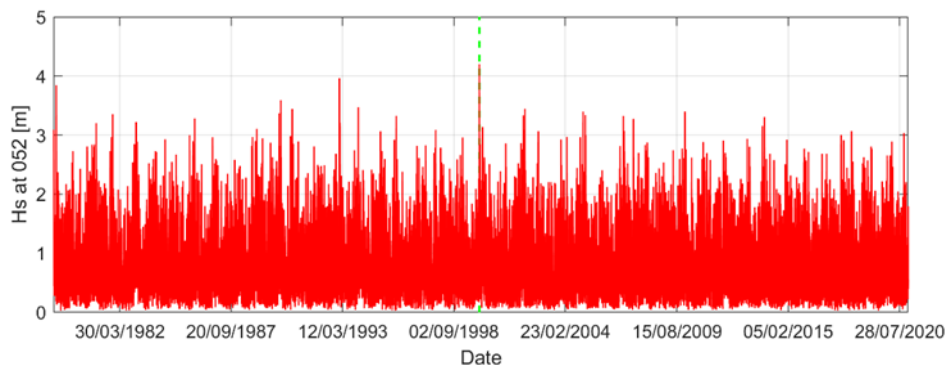


Figure 3.19 Timeseries of the computed significant wave height at the reference location in the Gennaker OWF over the period 1979-2021. The green dashed line indicates the timepoint of the extreme event selected to be modelled in the present study.

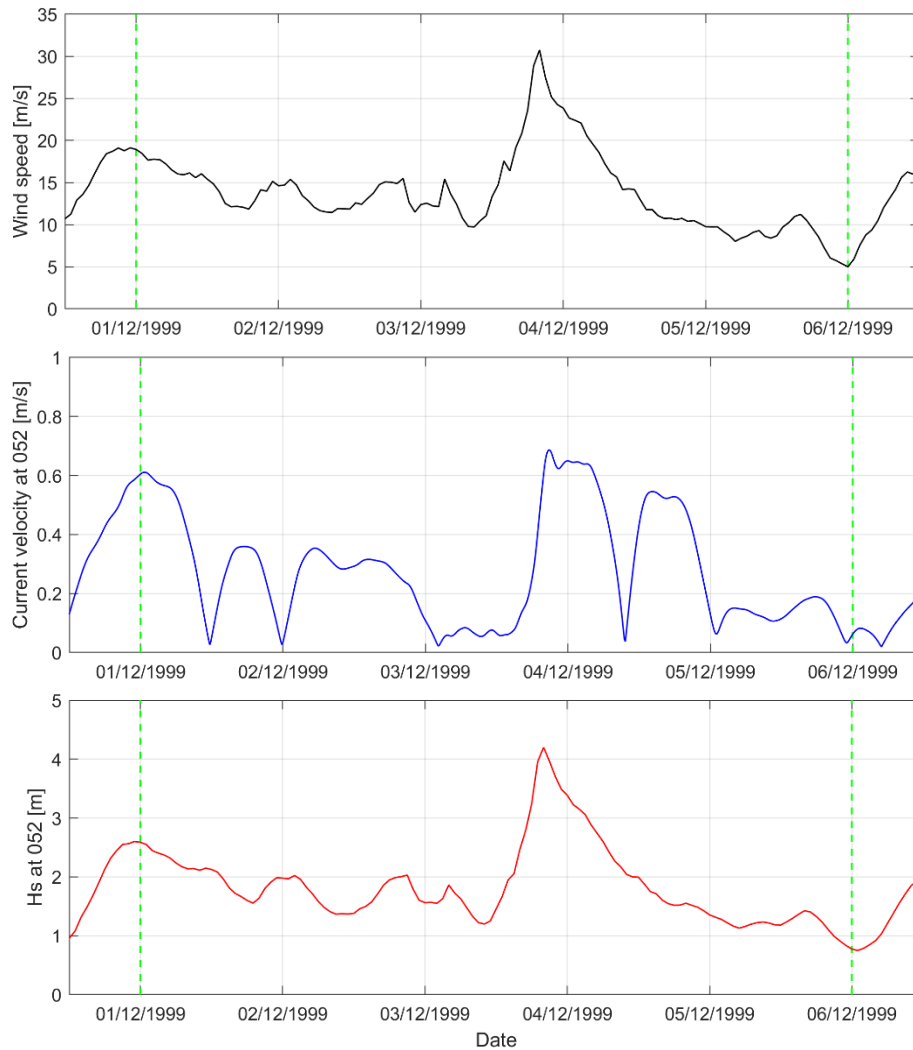


Figure 3.20 Timeseries of the ERA5 wind speed (top panel), simulated near-bottom velocities (middle panel) and simulated significant wave height (bottom panel) at the reference location in the Gennaker OWF during the occurrence of storm Anatol (01-Dec-1999 to 06-Dec-1999).

3.3.3.2 Model setup

Hydrodynamic model

For the hydrodynamic modelling the 2DH (depth-averaged) Dutch Continental Shelf Model (DCSM-FM), which has been extended to the Baltic Sea area (Deltares, 2022), is applied. An overview of the model extent both on a wider and regional scale can be found in Figure 3.21. The DCSM-FM is the sixth-generation hydrodynamic model, developed by Deltares for the Dutch Ministry of Infrastructure and Water Management (Rijkswaterstaat, RWS), and which has been extensively calibrated and validated (Zijl & Groenenboom, 2019). The extended DCSM-FM model used in the present study has been calibrated and validated using water level and velocity data in the surroundings of Gennaker OWF for several years (Deltares, 2022). The calibrated computational and physical parameter values of the model have been utilised in the present study. It is noted though that the additional scaling of the modelled velocities applied in the results of the metocean model has not been applied in the present study. This is counterbalanced by the amplification applied in the historical wind velocities (see next paragraph). In the end, the modelled peak depth-averaged velocity magnitude at the southwest edge of the Gennaker OWF during storm Anatol reaches and even exceeds the depth-averaged peak velocity associated with the 50-year significant wave height event reported at the nearby point 028 (Deltares, 2022).

The grid of the model has variable resolution that decreases with decreased water depth as well as with proximity to the area of interest, resulting to a grid cell size of 0.5 nm x 0.5 nm or approximately 840 x 930 m at Gennaker OWF. The bathymetry has been derived from the bathymetrical survey dataset from 2013-2015, supplemented by the EMODnet dataset (version 2020) as depicted in Figure 3.22. The model is forced with time and space varying meteorological conditions derived from ERA5, while at the open boundaries water levels consisting of a tide and surge component were provided. Wind velocities have been locally amplified, close to the peak of the storm, to increase the incident wave heights during Anatol storm to the 50-year significant wave heights (see also the wave model setup below). For detailed background information of the model including model setup, calibration and validation reference is made to report on the development of the sixth generation DCSM-FM (Zijl & Groenenboom, 2019) as well as the metocean study for Gennaker OWF (Deltares, 2022).

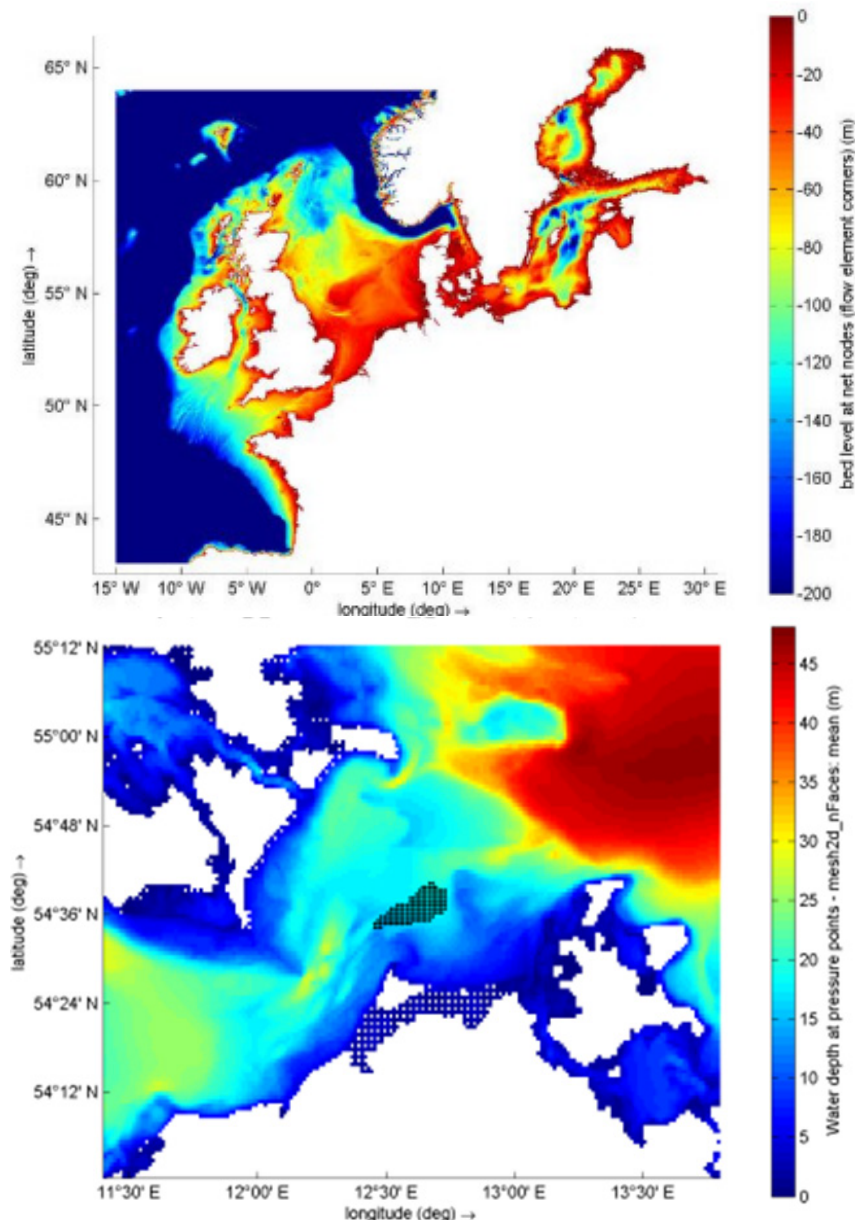


Figure 3.21 Bathymetry of the entire DCSM-FM model with Baltic extension (top panel) and local bathymetry (water depth) in the area surrounding the Gennaker OWF (the wind farm is marked by a black contour in the centre of the image) (bottom panel). Figures are taken from .Deltares (2022).

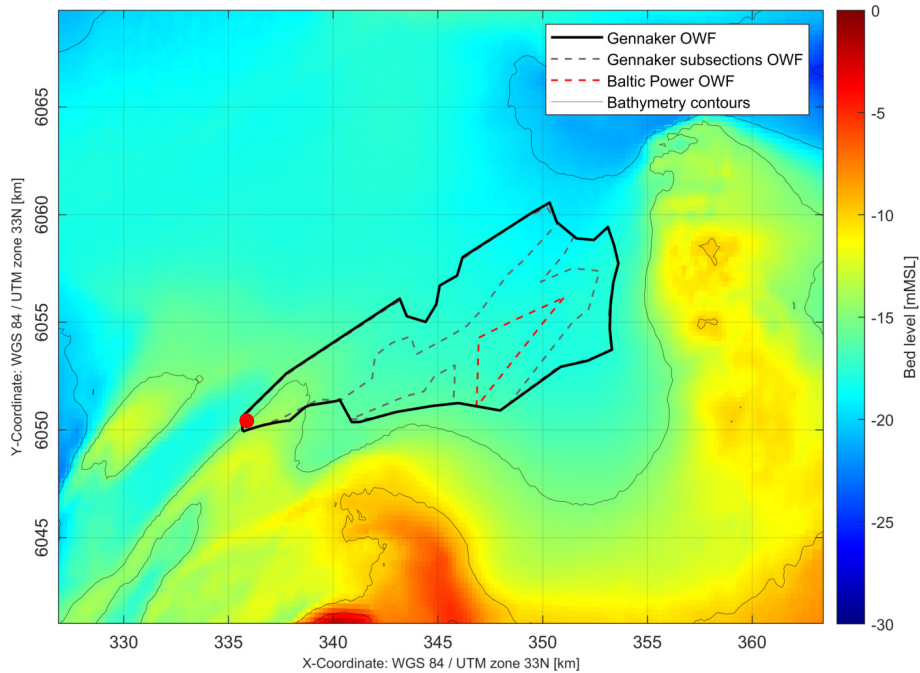


Figure 3.22 Bathymetry configuration used in the wave and flow numerical modelling. The red dot indicates the reference location used in this section.

Wave model

As a basis for the sediment transport modelling a detailed (D-WAVE) wave model is setup and coupled online to the hydrodynamic model. The D-WAVE model is based on the SWAN model setup used in the metocean study for the Gennaker OWF (Deltares, 2022). The model grid consists of 1 overall domain extending over the western part of the Baltic sea, similar to the extent of the SWAN model used in Deltares (2022). Two higher resolution domains extending over Gennaker OWF and the surrounding are nested in the overall domain, increasing the resolution to approximately 300 m at the area of interest as depicted in Figure 3.23. The grid with the highest resolution captures the variations in the local bottom gradients and depths in the area of interest.

Similar to the hydrodynamic model, the bathymetry information for the wave model was based on locally surveyed data from 2013-2015, supplemented by publicly available bathymetry data from the EMODnet dataset from 2020 as depicted in Figure 3.22. The wave model is coupled online to the hydrodynamic model. As such hydrodynamic information (water levels and currents) is passed at each timestep directly from the hydrodynamic model to the wave model. This means that the wave model accounts for the influence of the spatial distribution of water levels and currents (both speeds and directions) on wave propagation and evolution.

Numerical and physical parameters are defined based on the relevant settings for the metocean SWAN model (Deltares, 2022) or set to their default values (SWAN team, 2007). As part of the metocean study (Deltares, 2022), the SWAN model was calibrated and validated against available wave observations close to Gennaker OWF. It is noted that the additional scaling of the modelled wave heights applied in the results of the metocean model (Deltares, 2022) has not been applied in the D-Wave model for the present study. This leads to an underestimation of the modelled wave heights by the present model compared to those modelled in the metocean study (Deltares, 2022) and presented in Section 3.3.3.1. To correct for this underestimation of the modelled peak wave heights, wind speeds were locally amplified during the peak of the storm. As a result, for Anatol storm, the peak wave height simulated at the southwest edge of the OWF is 4.0 m, corresponding to a return period of 50 years as reported in Deltares (2022) for the nearby point 028.

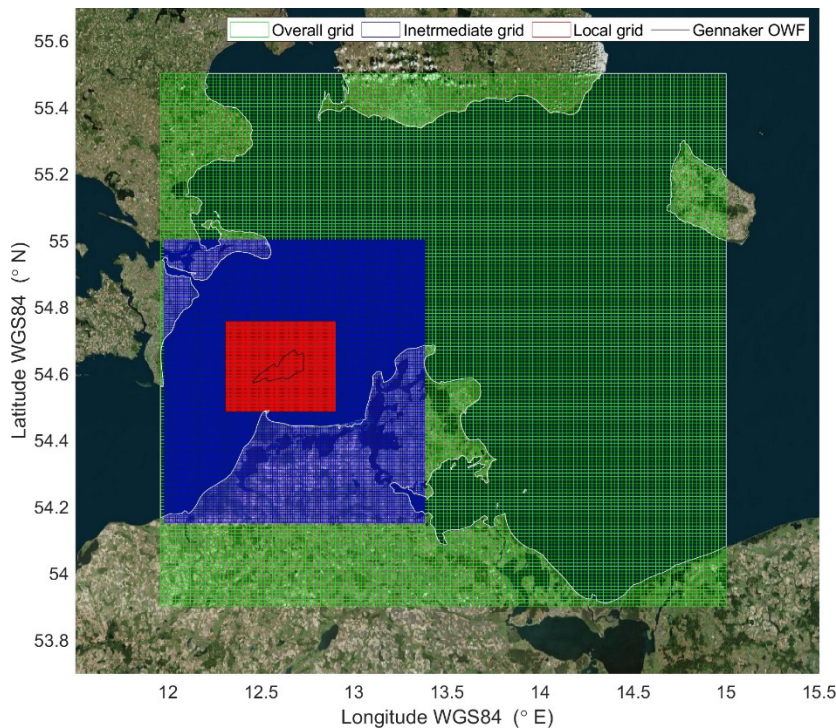


Figure 3.23 Gennaker wave model domains. The overall domain is marked with green color and the two nested domains, (intermediate and local) are blue and red respectively.

Sediment transport

The online sediment transport and morphology module of Delft3D-FM is used to simulate sediment transport throughout the hydrodynamic model domain at each computational time step (Lesser et al., 2004). The default settings, except for the sediment transport formula, have been applied in the sediment transport computation, which can be found in the D-Morphology User Manual (Deltares, 2020a). The TRANSPOR2004 transport equations (Deltares, 2020a) were used to model the movement of non-cohesive sand fractions and are implemented in the Delft3D flow solver. The Delft3D implementation of this formulation follows the principal description of Van Rijn (2007b, 2007a, 2007c) separating the sediment transport into suspended and bed-load components. Suspended sediment transport (balanced with erosion and deposition components from the seabed) is computed by the advection–diffusion equation and includes the effect of sediment in suspension on the fluid density. Bed load transports represent the transport of sand particles in the boundary layer in close contact with the bed surface and include an estimate of the effect of wave orbital velocity asymmetry. The built-in sediment transport model uses the Van Rijn (2007) sediment transport equations.

Conservatively, the bed was schematized to consist of a single sediment fraction with unlimited supply, with a median sediment diameter (D_{50}) of 100 μm which was deemed a conservative value, towards the lower end of the surficial sediment diameters present in the Gennaker OWF. Furthermore, such sediment may be cohesive (i.e., more resistant to erosion), but this effect is not considered in this model application. Note that the amount of geotechnical data for the Gennaker OWF is limited and the sediment diameter used in this simulation is an approximation from the data presented in Figure 2.7. Finally, bed level changes were enabled over time (allowing seabed levels to update during the simulation) to allow for the possible feedback between the waves/hydrodynamics and changing bed levels.

3.3.3.3 Model results

Figure 3.24, Figure 3.25 and Figure 3.26 present the simulated wave, flow and sediment transport conditions at one moment in time during peak conditions of storm Anatol (3-Dec-

1999). Close to Gennaker OWF both waves and current velocities are eastward & north-eastward directed, aligned with the wind direction. Within the OWF, wave heights during the peak of the storm range around 3.8 - 4.2 m, with the higher wave heights observed at the north side of the site. Depth-averaged current velocities range between 0.6 m/s and 1.2 m/s over the OWF during the peak of the storm. The highest velocities are observed at the southwestern area of the OWF where depths are shallower.

Figure 3.26 presents the simulated total (bed load and suspended) sediment transport conditions during the peak of the storm under the combined effect of waves and hydrodynamics. Similar to current flows, sediment transports are east to northeast directed. This is expected as typically currents are the main driver of transport of sediment and hence will dictate the transport directions. Waves predominantly increase erosion rates of the sediment, to be transported by the currents, by exerting an oscillatory shear stress on the seabed. Generally, the effects of waves near the seabed diminishes with increasing water depth and therefore becomes stronger in shallower waters. At those locations currents also appear stronger, as seen in Figure 3.25. As a result, peak sediment transports up to (55 m³/m/hr) appear at the shallower areas around Gennaker OWF.

Storm impact on the seabed is assessed cumulatively over the period 01-Dec-1999 to 06-Dec-1999 and is presented in Figure 3.26. This figure shows only local bed level differences in the order of some decimetres, after the storm, concentrated at the shallower areas south and east of Gennaker OWF. It is stressed that in these areas, more coarse sediment may be present (this is not investigated in this study, as the areas do not represent the area of interest), possibly resulting in lower changes in these areas in reality. The areas where bed level changes are observed are similar to the areas observed in the data-driven analysis as depicted in Figure 3.1.

Within the OWF, the simulated changes in the large-scale seabed (excluding the megaripples) are in the order of a few centimetres. This captures the variability of seabed levels in the Gennaker OWF during an extreme event with a return period of 50 years. Assuming occurrence of 1 such storm during the lifetime of Gennaker OWF and no seabed recovery, a maximum cumulative seabed change in the large-scale bathymetry of ± 0.07 m due to storm events is estimated and incorporated in the estimates of the future bed levels in Section 3.4 as a uniform value.

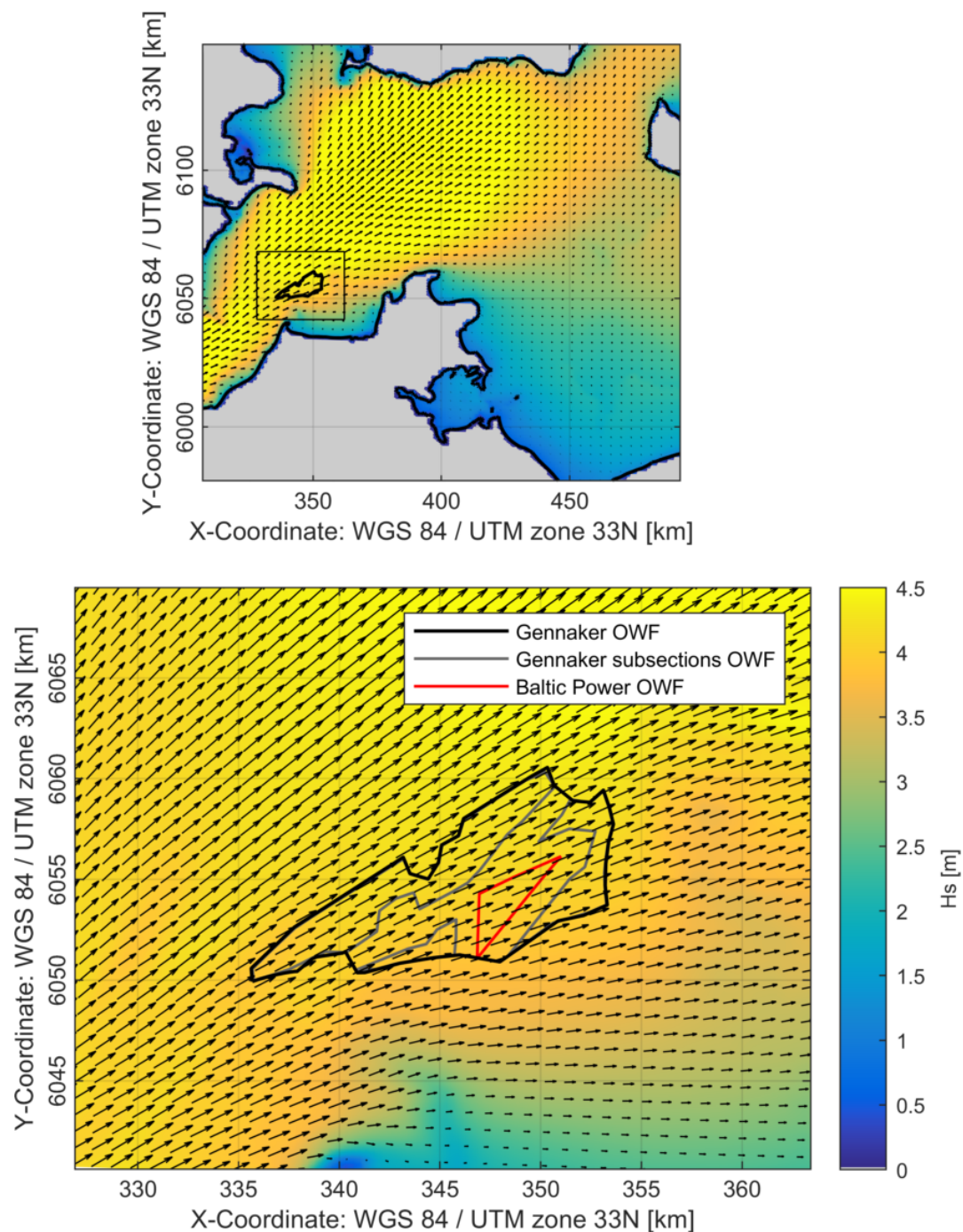


Figure 3.24 Simulated wave field during the peak of the storm (December 3, 1999, 20:00).

T = 03-Dec-1999 20:00:00

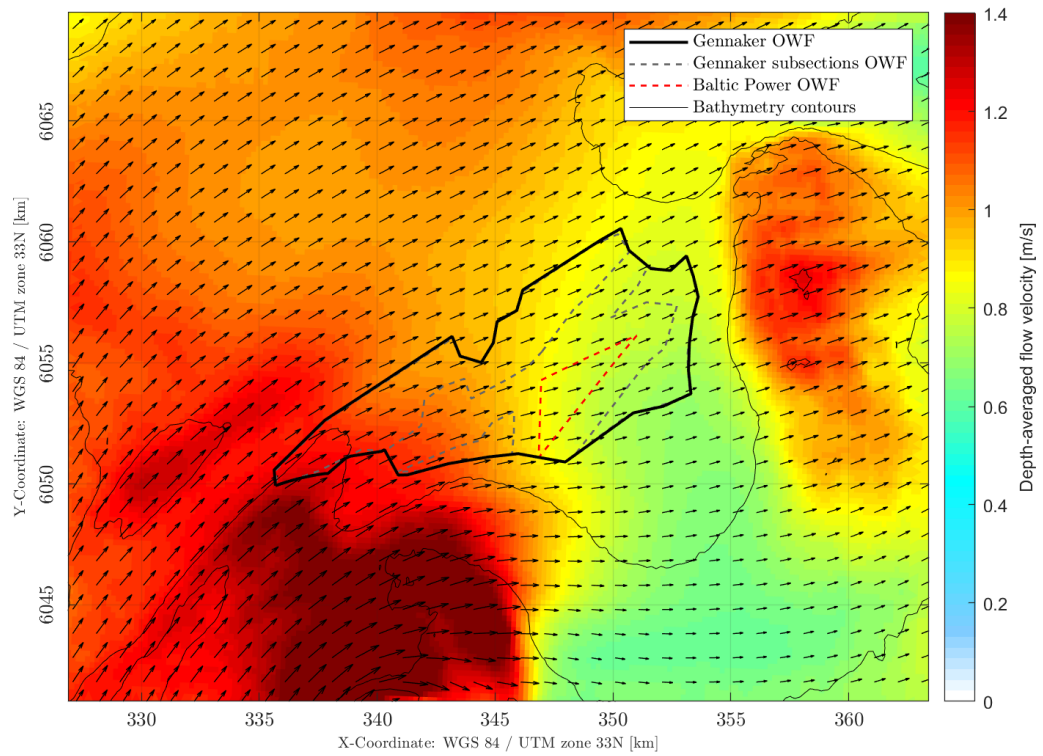


Figure 3.25 Simulated flow field during the peak of the storm (December 3, 1999, 20:00).

T = 03-Dec-1999 20:00:00

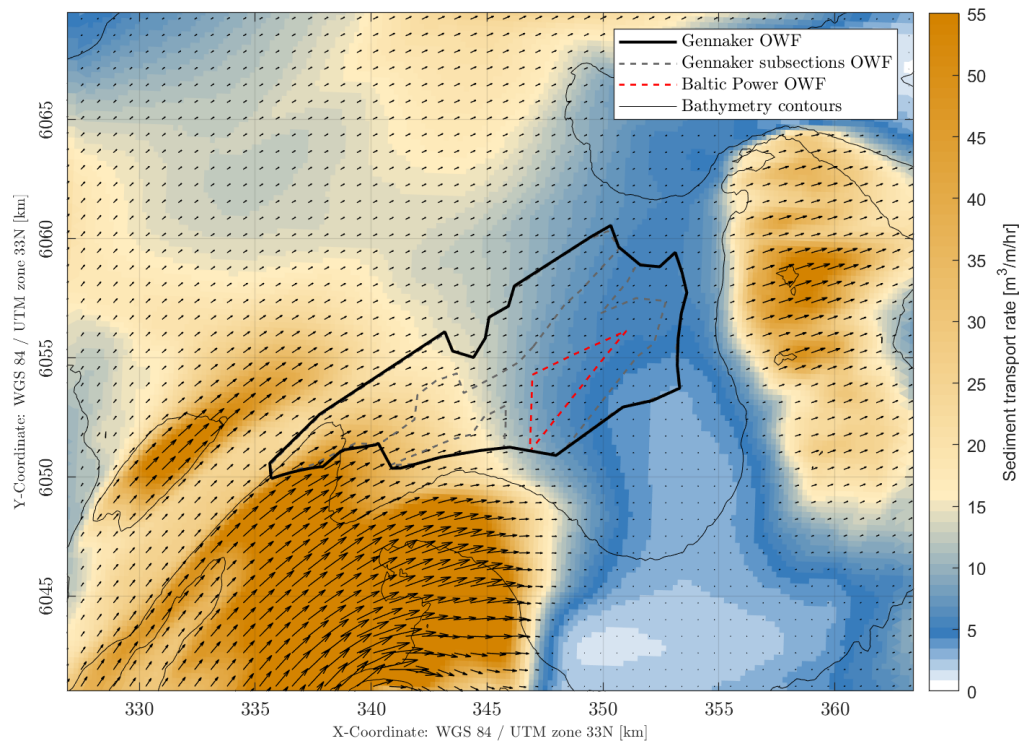


Figure 3.26 Simulated sediment transport rates during the peak of the storm (December 3, 1999, 20:00) using a median sediment diameter (D_{50}) of $100\ \mu\text{m}$.

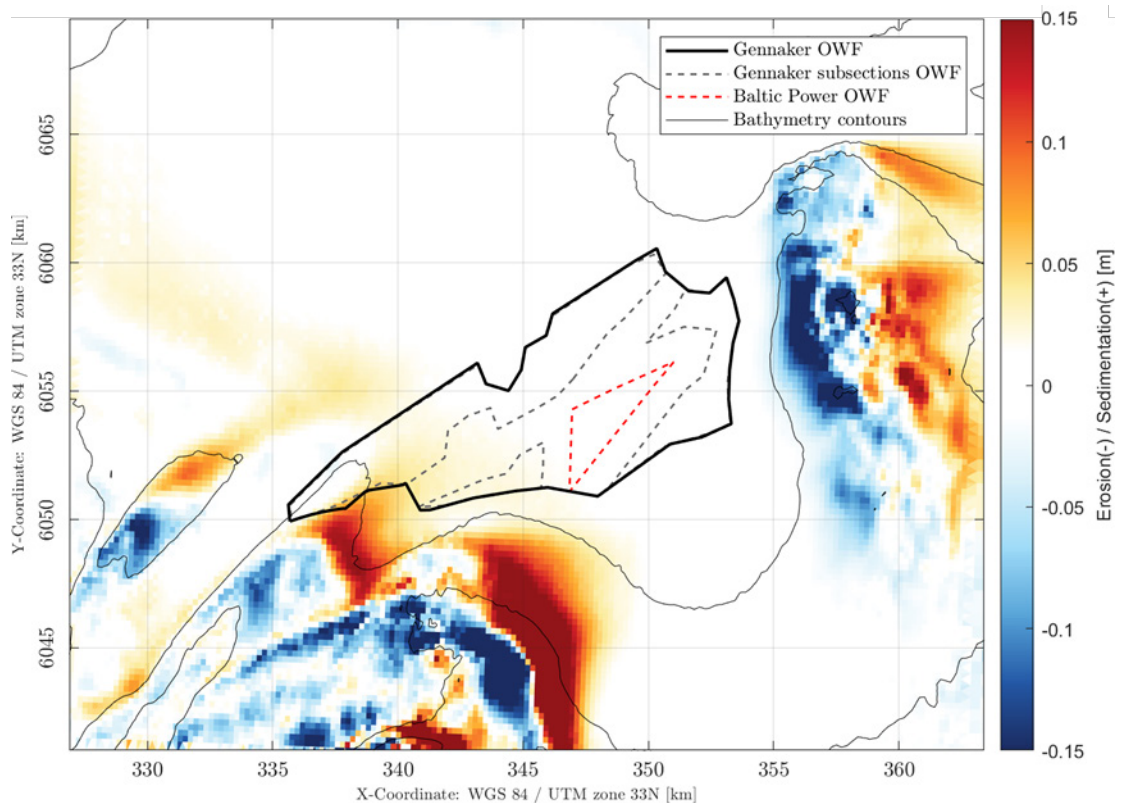


Figure 3.27 Simulated erosion(blue) and sedimentation(red) over the modelled period using a median sediment diameter (D50) of 100 μm .

3.4 Estimation of future bed levels

The morphodynamic analysis indicates that over the period 2013 to 2022 the mobility of the seabed in the Gennaker OWF is limited. After correction of the bathymetries (i.e., to correct for differences in vertical reference levels) the mobility of the seabed is estimated at maximum 0.15 m over a period of 5 to 7-years (i.e., from 2013 and 2015 to 2022). Analysis of notable areas showed that the largest seabed changes in the area (i.e., approximately ± 0.30 m) occur due to manmade cable repairs. The observed changes in seabed levels which can be attributed to natural morphodynamic activities are predominantly governed by the migration of small scale bedforms such as megaripples and uncertainties on seabed mobility around sediment structures.

Based on the morphodynamic analysis, it is expected that over the lifetime of the windfarm seabed mobility will be limited. However, small-scale seabed mobility cannot be fully excluded due to the megaripples located just south of the Gennaker OWF. It is unlikely that these bed features will suddenly form in the OWF, since the bed levels in the OWF are deeper. Nevertheless, although unlikely given the current water depths in Gennaker OWF, it cannot be guaranteed that these megaripples will not migrate into Gennaker OWF over the lifetime of the wind farm. It is stressed that this possibility is not included in the predicted future seabed levels.

It is also not possible to exclude large-scale seabed mobility based on the current morphodynamic analysis. The limited temporal spread of the bathymetric surveys (i.e., 6 years) and differences in vertical reference levels make it difficult to derive if any large-scale changes occur in the area. To account for seabed mobility caused by large-scale morphodynamics we thus consider a uniform uncertainty of ± 0.01 m per year over the full extent of the Gennaker OWF.

The extreme event assessment yielded limited seabed level changes inside the Gennaker OWF (i.e., <0.07 m). To capture changes caused by extreme events during the lifetime of the windfarm (i.e., changes during a 50-year storm event with a recurrence every 50 year), a uniform value of ± 0.07 m is considered in the future seabed levels.

It is noted that the observed seabed changes are influenced by data artefacts at the interface between track lines of the multibeam surveys and the differences in vertical reference levels. Moreover, recent multibeam measurements generally have a vertical measurement accuracy of approximately 0.20 m (International Hydrographic Organization, 2020). Most of the observed differences between the 2013-2015 and 2022 bathymetries fall within this uncertainty band.

Together, the uncertainties related to large-scale seabed morphology (± 0.01 m per year), the seabed level changes during an extreme event (± 0.07 m) and the uncertainties related to multibeam measurement (± 0.20 m) result in a maximum expected change of 0.66 m with respect to the Gennaker 2022 bathymetry at any given location over the lifetime of the windfarm (i.e., 4 years of preparation and installation till 2025/2026 and 35 years after installation). This ± 0.66 m uncertainty band might seem conservative based on the results of the morphodynamic analysis. However, the large- and small-scale morphodynamics may change over the lifetime of the windfarm.

The formulation for the predicted seabed level in any given year of this period is the following:

$$\text{Seabed level year } 20XX = \text{Seabed level } 2022 \pm (0.20 \text{ m (survey inaccuracy)} + 0.07 \text{ m (bed level changes during an extreme event)} + 0.01 \text{ m (yearly large-scale changes)} * 20XX - 2022)$$

Figure 3.28 shows the relative bed level change over the lifetime of the windfarm (i.e., 2022 to 2061) at any given location. The linear model consists of two components, the constant method uncertainty and morphology envelope. This model was used to determine the lowest seabed level (LSBL) and highest seabed level (HSBL) at the 103 Wind turbine Generators (WTG) locations and two Offshore Substations (OSS) locations. The LSBL and HSBL are calculated every year between 2022 and 2026 and every 5 years between 2026 and 2061. The results are provided in a separate Excel-file.

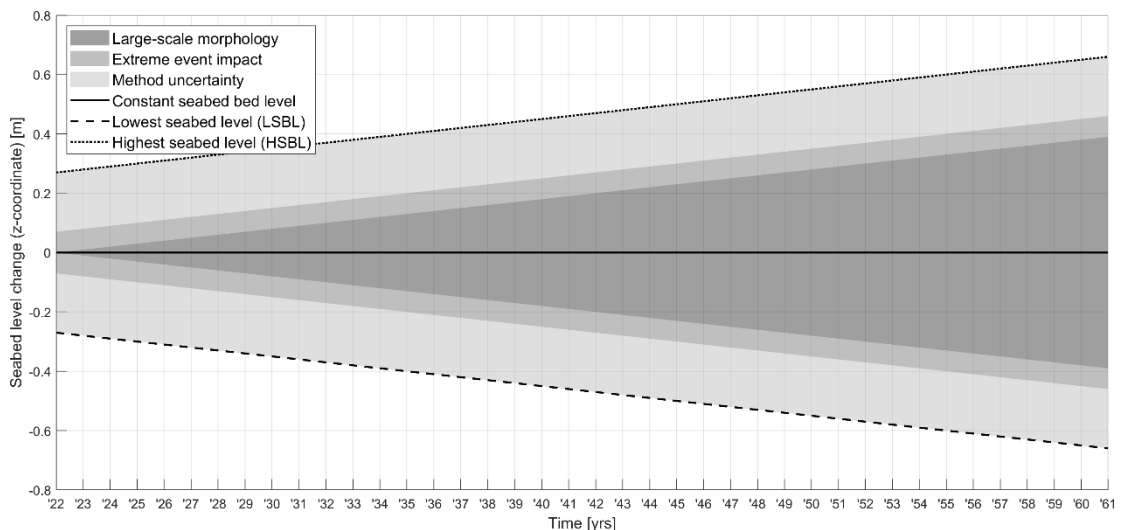


Figure 3.28 the relative bed level change over the lifetime of the windfarm (i.e., 2022 to 2061) at any given location with respect to the bathymetry survey conducted in 2022 by Vermessungsbüro Weigt (2022).

4 Conclusions and recommendations

4.1 Conclusions

Deltares has carried out a seabed mobility study for the Gennaker OWF. This is a proposed development by OWP Gennaker GmbH in the Baltic Sea approximately 15 km of the coast of Germany. The objectives of this study are to assess the morphodynamics and provide an estimation of the future seabed levels in and around the Gennaker OWF. To this end, a data-driven analysis is performed that consist of three parts: overview of the characteristics in the Gennaker OWF, morphodynamic analysis and estimation of future bed levels.

The hydrodynamics and wave data showed that the hydrodynamic conditions in the area are mild. The geotechnical and geophysical data illustrated that the substrate in and around the Gennaker is generally characterised by very fine to fine sands. The morphodynamic analysis indicated that over the period 2013 to 2022 the mobility of the seabed in the Gennaker OWF is limited. Similar conclusions were drawn from the extreme event assessment. Numerical modelling of the Anatol storm showed limited seabed level changes (<0.02 m) inside the Gennaker OWF. Nevertheless, seabed mobility cannot be fully excluded due to the limited temporal spread of the bathymetric surveys (i.e., 6 years), limited soil information and uncertainties related to multibeam measurements.

To account for seabed changes caused by small-scale dynamics (i.e., migration of megaripples) and large-scale dynamics (i.e., regional seabed changes) we thus consider a uniform uncertainty of ± 0.01 m per year over the full extent of the Gennaker OWF. For the extreme events a uniform value of ± 0.07 m was considered, taking into account both the results of the extreme event assessment and the recurrence of such an event. Together with the uncertainties related to multibeam measurement (± 0.20 m) a maximum change of ± 0.66 m with respect to the Gennaker 2022 bathymetry is expected at any given location over the lifetime of the windfarm (i.e., 4 years of preparation and installation till 2025/2026 and 35 years after installation).

4.2 Recommendations

Although it is concluded that there is little potential for seabed mobility in this region, there was limited availability of bathymetry geotechnical data (e.g., at the foundation locations) to further confirm this. Additional geotechnical data would be of benefit for the assessment of local morphodynamics, i.e., scour, around foundations, as well as further investigations into the sediment structures observed in the seabed survey.

It is also advised to further expand the current available bathymetry datasets over time (during installation), to confirm the anticipated behaviour as described in this report. An additional bathymetry dataset provides further confirmation on the offset in the 2013-2015 bathymetries, a third point (or fourth if considering the older BSH data) in time to increase the temporal and spatial resolution of the available data (i.e., 2013 till 2025/2026 when conducting prior to installation) and a starting point during installation of the wind farm. It is stressed that for the latter also bathymetry surveys at the locations of the foundations could be conducted. In case a full area survey is executed it is strongly advised to also incorporate the shallow area southwest of Gennaker OWF.

During wind farm operation, it is advised to monitor the seabed around each foundation periodically (e.g., every two years) or following extreme events (e.g. after the occurrence of an

extreme event). This way, potential risks related to seabed mobility can be identified and mitigated beforehand.

In addition, it is advised to address the potential effects of wind farm related infrastructure in relation to the observed trends in this study. Although in rare occasions and not in large quantities, there is a clear surge-driven, net sediment transport direction which could lead to sediment infill of e.g., a cable trench.

Generally, we expect the impact of the wind farm related infrastructure (e.g., wind turbines, cables) on the ambient flow to be small in absolute terms, since background flow velocities and sediment transports under normal conditions are already small, though this also depends on the design dimensions. From our present understanding of the system, we do therefore not expect the potential scour around these types of infrastructure to be high, but that would need to be further studied.

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A Hydrodynamic and wave conditions

This appendix presents the hydrodynamic and wave conditions in the Gennaker OWF as depicted by roses for locations 028, 101 and 140 in Deltares (2022). Locations are depicted in Figure 2.1.

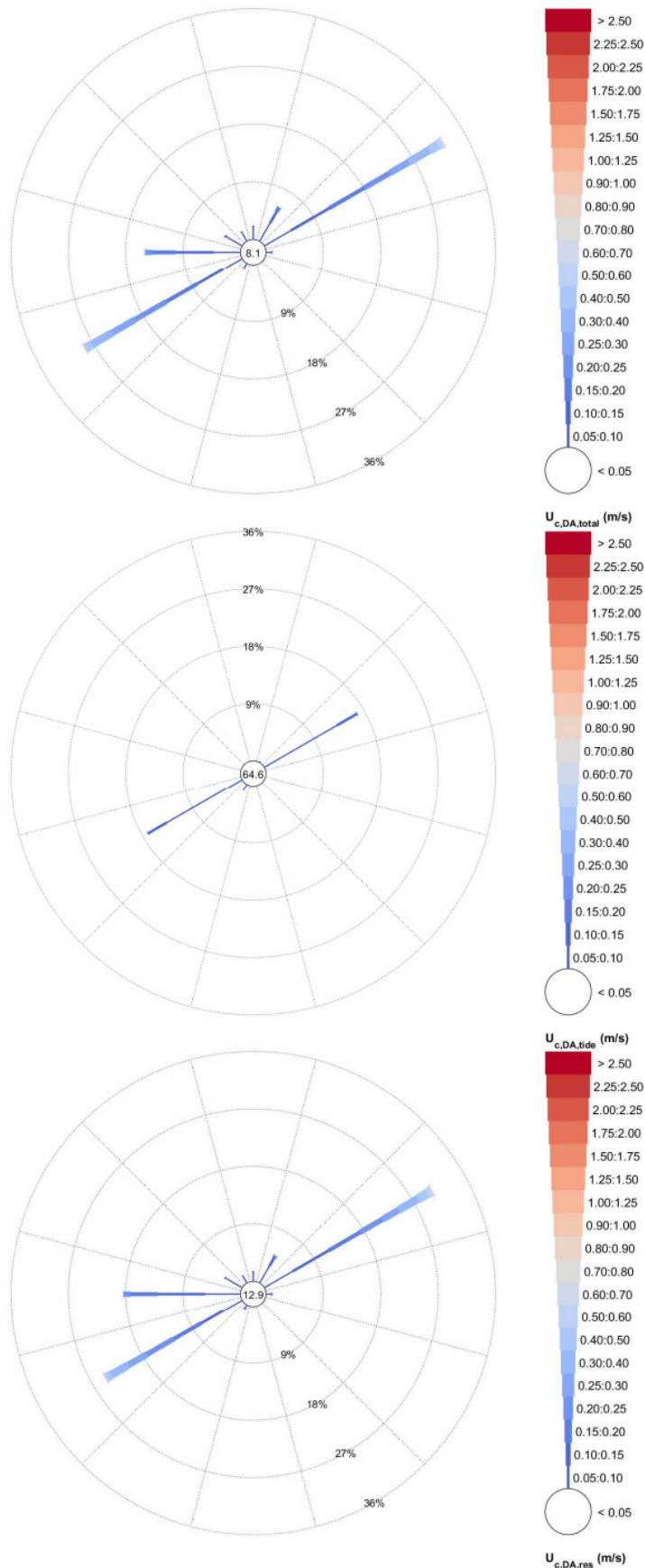


Figure A.1 Roses of the depth-average total (top panel), tidal (middle) and residual (bottom panel) current speed and direction for normal conditions at location 028 (Deltares, 2022). Location is depicted in Figure 2.1.

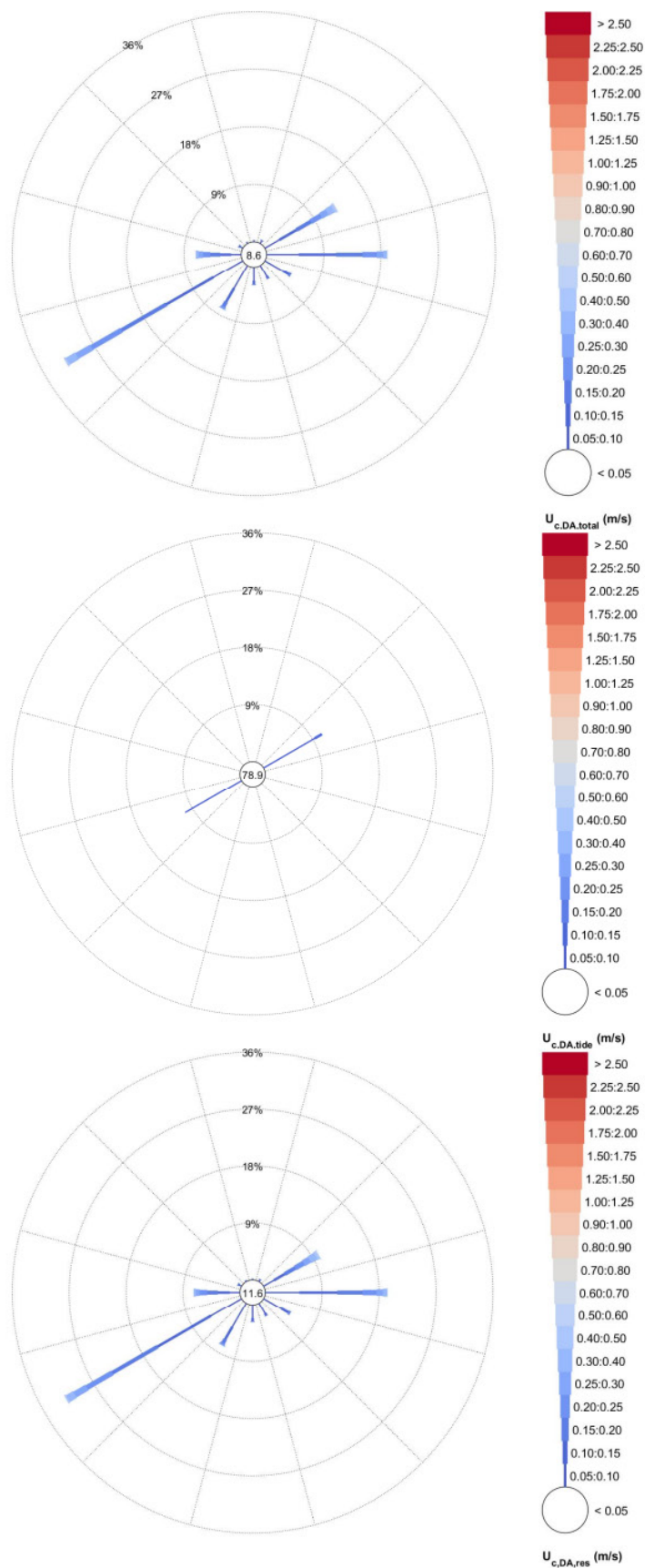


Figure A.2 Roses of the depth-average total (top panel), tidal (middle) and residual (bottom panel) current speed and direction for normal conditions at location 101 (Deltares, 2022). Location is depicted in Figure 2.1.

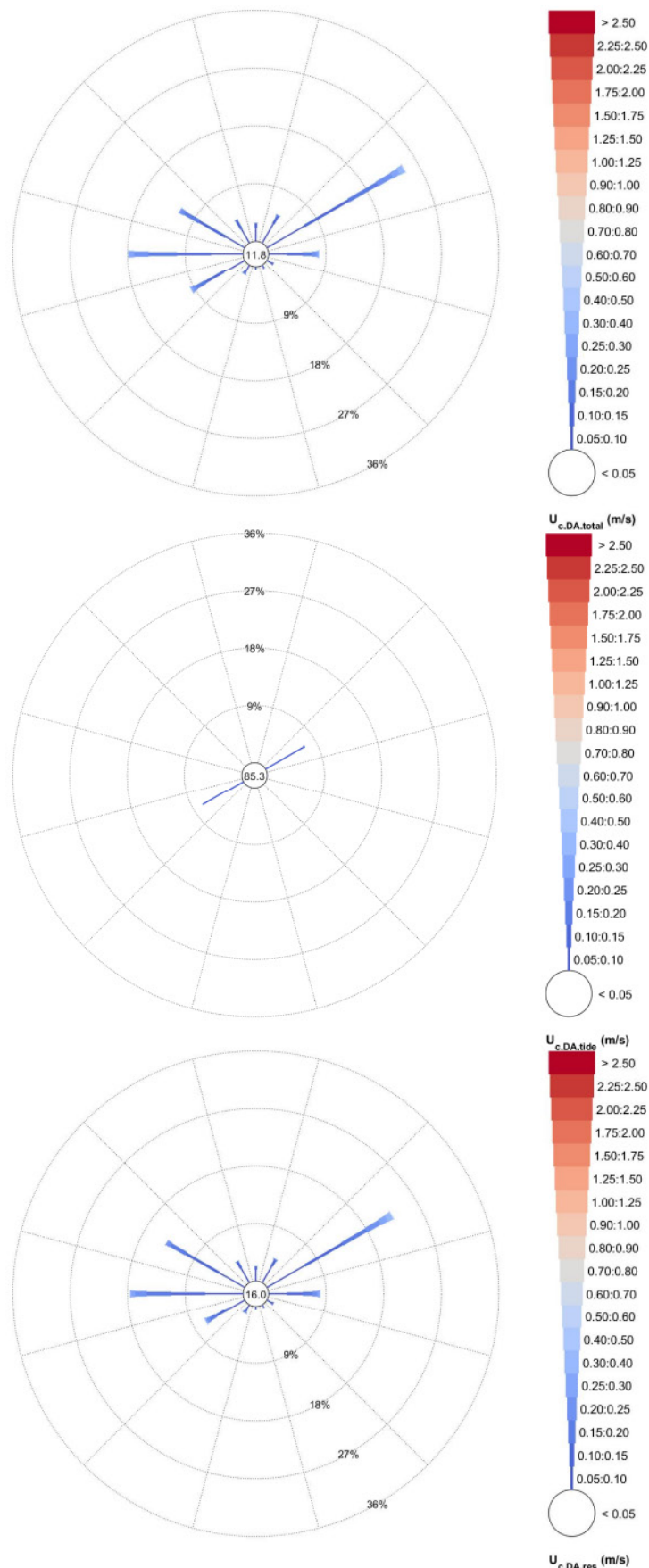


Figure A.3 Roses of the depth-average total (top panel), tidal (middle) and residual (bottom panel) current speed and direction for normal conditions at location 140 (Deltares, 2022). Location is depicted in Figure 2.1.

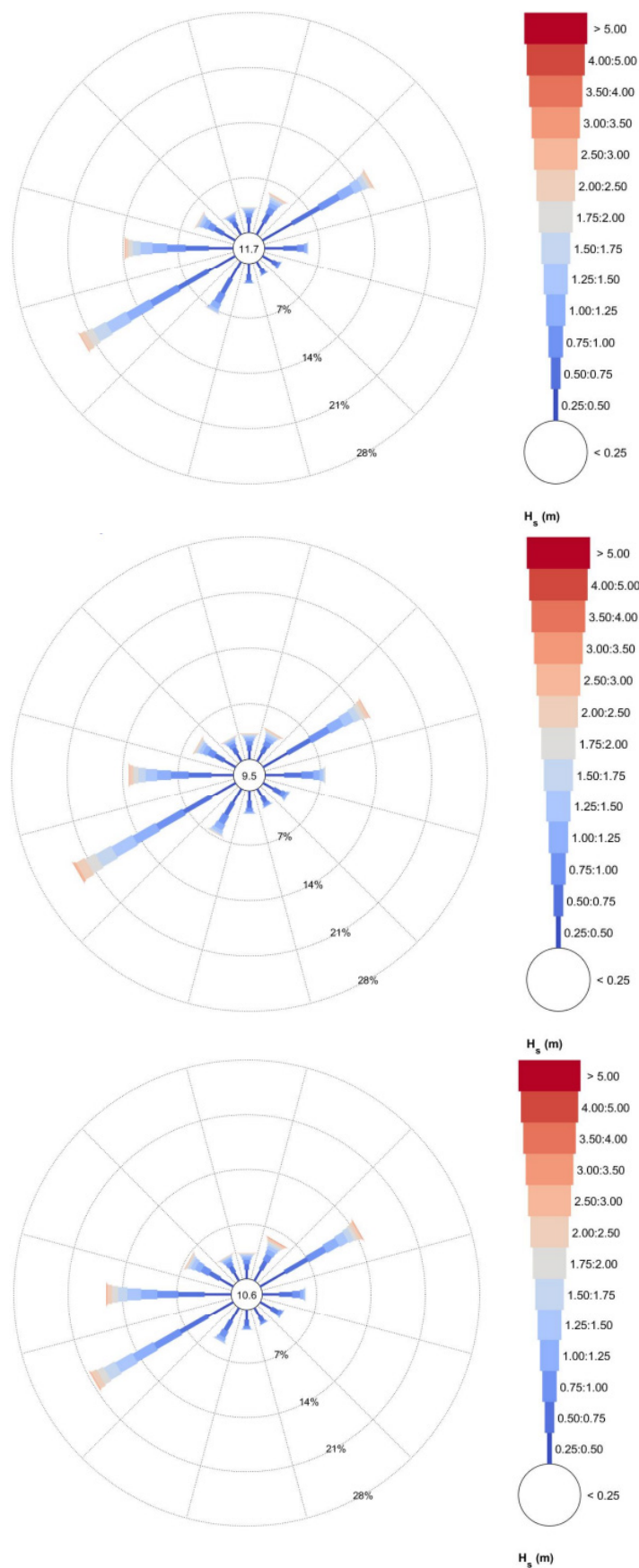


Figure A.4 Roses for the significant wave height and direction for normal conditions at location 028 (left), 101 (middle) and 140 (right) (Deltares, 2022).

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